

Explosive Diversification: Cartel Adaptation to Crackdowns in Mexico*

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Abstract

Oil theft from pipelines funds armed criminal groups around the world. In Mexico, cartels first diversified into oil theft following a crackdown on drug trafficking, then diversified again into gas theft after a crackdown on oil theft in 2019. We develop a model of criminal diversification in which sector-specific capital (i.e., technical skills, corrupted officials, territorial networks) is durable and partially transferable to adjacent sectors. When a crackdown raises costs in one sector, previously accumulated capital facilitates entry into others, irreversibly expanding the criminal opportunity set. Consistent with this ratchet mechanism, we show that Mexico's 2019 crackdown on oil theft was associated with a 1,583% increase in gas pipeline thefts (from near-zero at baseline), an 18% increase in cartel presence, and a 19% increase in homicide rates in gas pipeline municipalities, with effects concentrated near oil infrastructure and driven most strongly by oil theft-specialized cartels. After the crackdown, gas theft persisted while oil theft bounced back, indicating durable expansion of cartel capacity. Findings reveal path-dependent criminal adaptation to enforcement that endogenously expands the reach and violence of organized crime.

JEL codes: K42, Q34, Q35, O17

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1 Introduction

Crackdowns on organized crime often displace rather than eliminate criminal activity. A growing literature documents how enforcement targeting one illicit sector prompts criminal organizations to diversify into less-policed alternatives, generating unintended spillovers of crime and violence (Battiston et al., 2024; López Cruz and Torrens, 2023; Dell, 2015; Parker and Vadheim, 2017; Chimeli and Soares, 2017). Combating this diversification is an urgent policy challenge, given that organized crime increases violence (Sobrinho, 2020), reduces economic growth (Fenizia and Saggio, 2024), provokes forced migration (Daniele et al., 2023), and corrupts governance and elections (Trudeau, 2024; Lessing, 2018).¹

Previous studies have largely examined episodes of criminal diversification in isolation, without exploring how sequential, sector-specific law enforcement efforts may dynamically ratchet up the capabilities of criminal groups. We develop a model of this ratchet effect and analyze Mexico’s 2019 crackdown on oil pipeline theft as an empirical test of model predictions. Following Mexico’s War on Drugs in the mid-2000s, cartels diversified from drug trafficking into oil pipeline theft (Battiston et al., 2024; López Cruz and Torrens, 2023), and then diversified again into gas pipeline theft – a more technologically challenging and dangerous activity – following a subsequent crackdown on oil theft in 2019.² After each enforcement shock, cartels continued to operate in the targeted sector while expanding into a new one, accumulating organizational capacity with each adaptation. This sequential expansion, from drugs to oil to gas, is the ratchet dynamic that our model formalizes.

Hydrocarbon theft from pipelines is itself a growing global problem: worldwide, oil thefts exceeded 200 million barrels and US\$11 billion in 2015, funding armed groups across Colombia, Ecuador, Indonesia, Mexico, Nigeria, Russia, and Syria (Valencia, 2024, 2022; Rexer and Hvinden, 2026; May, 2017). Until recently, theft of natural gas and liquefied petroleum gas (LPG) from pipelines was far less common, but Mexico’s experience, which we document here, presages emerging risks elsewhere. Gas pipeline infrastructure is expanding rapidly as countries shift from coal to gas for electricity generation and households adopt gas for cooking and heating

¹Across Latin America, between 77-101 million people – 14% of the population – live under the influence of organized criminal groups (Uribe et al., 2025). This makes organized crime one of the most severe threats to governance and development in the region (Corrales and Freeman, 2024). Mexico is particularly impacted by violence associated with organized criminal groups (i.e., cartels), whose presence in communities reduces profits and salaries and increases poverty and unemployment (Gutiérrez-Romero and Oviedo, 2017).

²In Mexico, oil theft involves tapping pipelines to siphon refined oil products such as gasoline and diesel. This activity is dominated by cartels, which stole an average of 58 thousand barrels of gasoline per day, amounting to US\$3.5 billion in value and 8% of national consumption in 2018 (Solis, 2018). When prices are high, a cartel can siphon US\$90,000 worth of gasoline from a pipeline tap in just 6 minutes (May, 2017). Cartel diversification toward gas theft after 2018 received contemporaneous media attention (Nájar, 2019) and corresponded with a sharp increase in gas-related accidents, from an annual average of 36 between 2003-2017 to 275 in 2019 and 307 in 2020 – 63% of which involved fires or explosions (CENAPRED, 2021).

(Kojima, 2021; International Energy Agency, 2019). In 2024, there were over 1 million km of gas pipelines operating globally and 238,000 km proposed or under construction – 81% of these in low- or middle-income countries (Global Energy Monitor, 2025).

We first develop a model of sequential criminal adaptation in response to enforcement crackdowns. A profit-maximizing cartel allocates limited organizational capacity across a portfolio of illicit sectors, accumulating durable sector-specific capital (e.g., technical expertise, corrupted officials, territorial networks) that is partially transferable to adjacent sectors. When a crackdown increases the costs of operating in one sector, capital transferability reduces entry costs into related sectors, tipping the cartel toward diversification. The model generates a ratchet effect in which each diversification event lowers barriers to further entry through accumulated expertise and expanded organizational capacity. Sectors are entered sequentially by proximity to the cartel’s existing capital stock, and diversification persists even when enforcement in the originally targeted sector eases. The model delivers several empirical predictions. Gas theft should rise following the oil crackdown, concentrated in municipalities neighboring or hosting oil pipelines and driven disproportionately by cartels with prior oil theft experience. Gas theft should persist even when oil enforcement eases, as accumulated capital is durable and entry costs are sunk. Finally, homicides should rise in gas pipeline municipalities as diversification generates encroachment into rival territories and coercion of officials and operators.

We test these predictions in the context of Mexico’s crackdown on oil theft, which involved the shutdown of high-theft oil pipelines between December 2018 and April 2019, temporary transportation of fuel by truck during the closure, and increased military policing along pipeline corridors. We combine georeferenced maps of refined oil and gas pipelines with annual municipality-level reports of pipeline thefts, cartel presence, and homicides. We use difference-in-differences and event study specifications to measure changes in these outcomes before and after the crackdown in two treated groups: directly targeted municipalities hosting refined oil pipelines, and indirectly exposed municipalities hosting gas pipelines. While oil pipeline municipalities were endogenously targeted due to worsening thefts and violence (we interpret results for this group as descriptive), gas pipeline municipalities received a plausibly exogenous shock – they were not specifically targeted by the crackdown, yet experienced spillovers as cartels displaced from oil theft diversified into gas. Both treated groups are compared to control municipalities that host no pipelines. We additionally examine effects on other crimes and spatial spillovers of violence into neighboring non-pipeline municipalities.

Results show that the 2019 crackdown was associated with a large reduction in illegal taps on refined oil pipelines in the short term, reversing a trend of exponential growth since 2010.

Simultaneously, however, the crackdown resulted in a 1,583% increase in LPG thefts from gas pipelines (+3.8 taps per municipality-year from a baseline mean of just 0.24). In line with the model’s spatial channel, this increase was concentrated in municipalities that neighbor or host oil pipelines, indicating that cartels displaced from oil theft shifted spatially toward nearby gas infrastructure. In line with the technological channel, increased cartel activity in gas pipeline municipalities was driven most strongly by cartels with previous experience in oil theft, though presence of cartels without prior oil theft experience also grew. When the oil theft crackdown eased after April 2019, oil theft began to rise again and recovered to 80% of its 2018 peak by 2023, while gas theft remained elevated – consistent with the partial displacement and persistence mechanisms of our model.

Cartel diversification into gas theft was accompanied by a sharp increase in violence. Homicide rates in gas pipeline municipalities increased by 19% (+4 per 100,000 residents) following the crackdown, with increases concentrated among men (the most likely cartel participants and victims), in municipalities neighboring or hosting oil pipelines, and in places where cartels were absent before the crackdown but subsequently entered. In directly targeted oil pipeline municipalities, the crackdown failed to root out violence or cartel presence: homicide rates remained unchanged at levels well above the national average, and the number of active cartels *increased* by an average of 1. These findings are consistent with the ratchet effect: rather than eliminating criminal activity, the 2019 crackdown on oil theft expanded it, leaving cartels operating across a wider geographic footprint and broader range of illicit sectors than before.

1.1 Related Literature and Contributions

This study contributes to three strands of literature on the economics of crime: (i) the behavior of organized criminal groups, (ii) unintended consequences of enforcement and crackdowns, and (iii) crime focused on extractive industries, particularly pipeline theft.

Literature on organized criminal groups has shown that they infiltrate and loot healthy firms (Mirenda et al., 2022), prey on profitable farmers (De Haro, 2025), undermine free and fair elections (Trudeau, 2024), and reduce economic growth (Fenizia and Saggio, 2024). Governments face trade-offs in combating these groups, as targeting their revenue sources may lead them to prey on local residents instead (Blattman et al., 2024). López Cruz and Torrens (2026) show that criminal organizations also adapt to external demand shocks, documenting how surging US fentanyl demand from 2013 onward prompted Mexican trafficking organizations to re-optimize their routes, spreading violence to new areas. We build on this literature by developing a formal model of sequential criminal adaptation and studying cartel diversification in response to

a major crackdown using rich data on cartel presence and crime. Our model and results reveal that cartel diversification into gas theft aligns with prior patterns of specialization, with cartels behaving as firms that accumulate durable skills and territorial networks in specific criminal sectors and face both spatial and technological barriers to entry into new ones.

Studies of law enforcement crackdowns have largely concluded that criminal groups are adaptable and resilient, and that disruptions of illicit equilibria increase violence (Magaloni et al., 2020; Herrera and Martinez-Alvarez, 2022; Jones, 2013). For instance, a policy aimed at preventing trade of conflict minerals from the Democratic Republic of the Congo led warlords to shift toward gold mining and looting of civilians (Parker and Vadheim, 2017). In the Brazilian Amazon, the prohibition of mahogany harvesting led to higher violence without reducing the mahogany trade (Chimeli and Soares, 2017). Finally, Dell (2015) finds that crackdowns on drug trafficking in Mexico increased violence as cartels fragmented and took advantage of weakened rivals. We contribute to this literature by going beyond individual episodes of diversification to formally model how repeated enforcement shocks ratchet up criminal organizational capacity, and by validating these predictions in the context of Mexico’s 2019 crackdown on oil theft, which triggered violent cartel diversification into a new and explosive activity: gas pipeline theft.

Extractive industries are especially vulnerable to criminal appropriation because they present fixed point targets and generate large rents (Dube and Vargas, 2013; Angrist and Kugler, 2008). Berman et al. (2017) show that increases in world mineral and metal prices significantly increase violence around African mines as groups fight to control resource windfalls. Our findings provide further evidence that fixed extractive infrastructures (i.e., pipelines) are vulnerable to criminal appropriation and associated violence, even far removed from active production areas.

Most specifically, we build on existing studies of pipeline theft, which presents a growing problem in countries including Brazil, Colombia, Ecuador, Indonesia, Mexico, Nigeria, Russia, and Syria (Esberg, 2025; Franco-Vivanco et al., 2023; Ventriglia et al., 2023; Rexer, 2023). López Cruz and Torrens (2023) document increased oil thefts and violence near oil pipelines after a government crackdown on drug trafficking in Mexico in 2007, providing evidence that cartels substituted for lost drug revenues with fuel theft. Battiston et al. (2024) leverage the same 2006 drug trafficking crackdown to show that upstart cartels leapfrogged incumbents to engage in pipeline thefts, leading to cartel specialization and segmentation of illegal markets. We go beyond previous studies of oil pipeline theft in Mexico by documenting a new stage in the story of cartel diversification – from oil thefts to gas – which represents a significant increase in technical sophistication and infiltration of state agencies. More broadly, we contribute to the literature on global pipeline theft by quantifying the emergence of gas theft at scale for the first

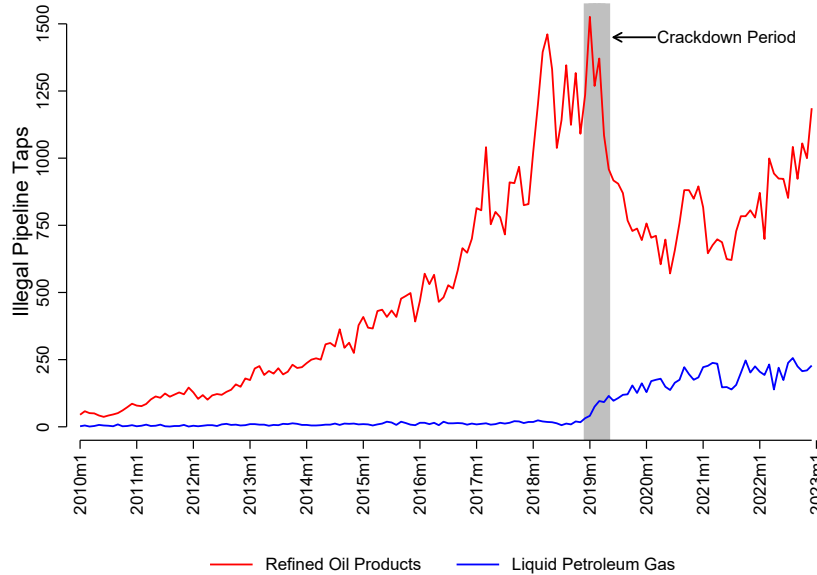
time. The growth of this lucrative and explosive market in Mexico presages threats to other gas-producing regions around the world.

2 Background

2.1 Mexico’s War on Drugs and the Rise of Pipeline Theft

In 2006, Mexico’s government launched a War on Drugs to combat drug trafficking organizations. The government employed a kingpin strategy, which involved apprehending leaders of these organizations. This tactic destabilized large drug cartels, leading to their fragmentation, increased competition from newly formed organizations (Calderón et al., 2015), and diversification into non-drug-related activities, including extortion of agricultural workers (De Haro, 2025) and theft of refined oil products from pipelines – with accompanying increases in violence (Battiston et al., 2024; López Cruz and Torrens, 2023). Reports of illegal taps on refined oil pipelines increased dramatically following the War on Drugs, peaking at 15,000 reported taps in 2018. Figure 1 shows monthly reports of illegal taps on refined oil and gas products between 2010 and 2022.

Figure 1: Illegal Taps on Refined Oil and Gas Pipelines



Note: Data were obtained from Pemex, Mexico’s state-owned petroleum company, through a freedom of information request. Refined oil products are defined as gasoline and diesel.

Criminal organizations have developed effective methods to steal refined oil products (primarily gasoline and diesel) from pipelines operated primarily by Pemex, the state-owned oil company.³ Cartel members pay or coerce pipeline operators to share information on pipeline

³In 2013, a nationwide energy reform allowed private investments in crude oil exploration, as well as gas and

locations and provide the necessary equipment and information on how to extract the fuel (Cultura Colectiva, 2019; López Cruz and Torrens, 2023). Collaborating pipeline operators also notify cartels of the best time to drill to avoid an explosion, and of the type of fuel (e.g., gasoline or diesel) passing through the pipeline. The stolen fuel is loaded into barrels or trucks and sold in local markets or along local roads. Consumers of stolen fuel products include factories, transportation businesses, taxi drivers, and gas stations. Fuel theft can also subsidize local fuel prices and constitutes a valuable source of income for local residents, complicating efforts to combat this lucrative activity (Ralby, 2017; Torres, 2017). In 2018, Pemex reported losses of approximately US\$3.5 billion due to illegal gasoline taps, which amounted to 8% of national gasoline consumption (Solis, 2018). Fuel theft also causes substantial losses in tax revenues, which Mexico’s tax authority estimated at US\$3.15 billion in 2021 (Tapia Cervantes, 2021).

As illustrated in Figure 1, thefts of liquid petroleum gas (LPG) from gas pipelines were almost unheard of prior to 2019. Corresponding with the crackdown on refined oil theft beginning in December 2018, illegal gas taps quickly increased by over 500%. Approximately 50,000 tons of LPG were stolen monthly in 2022 (Usila, 2022) – accounting for 7.6% of national consumption. LPG presents unique opportunities for criminal organizations because 79% of Mexican households use it for cooking and heating (INEGI, 2022).

Stolen LPG is stored in small cylinders and sold to local consumers informally. Gas pipeline theft is intrinsically more dangerous than oil pipeline theft, since gas is stored under extreme temperatures and pressures and taps cannot be made while the pipeline is in use. Instead, thieves rely on tip-offs on when the pipeline will be available for tapping, or provoke accidents that require pipes to be shut down (Nájjar, 2019). The rise in gas pipeline theft is thus closely associated with a dramatic rise in gas-related accidents (mostly fires and explosions), from an average of 36 per year between 2003 and 2017 to 275 in 2019 and 307 in 2020 (Appendix Figure A1). Gas thefts are often undertaken in densely populated urban areas, putting local residents at risk. Mexico’s National Center for the Prevention of Disasters reports 208 deaths and 2,352 injuries from gas fires and explosions since 2003, with most occurring in recent years (CENAPRED, 2021).⁴

2.2 The 2019 Crackdown on Oil Pipeline Theft

Rising levels of oil pipeline theft and associated violence reached a boiling point in 2018, corresponding with the beginning of Andrés Manuel López Obrador’s term as president of Mexico.

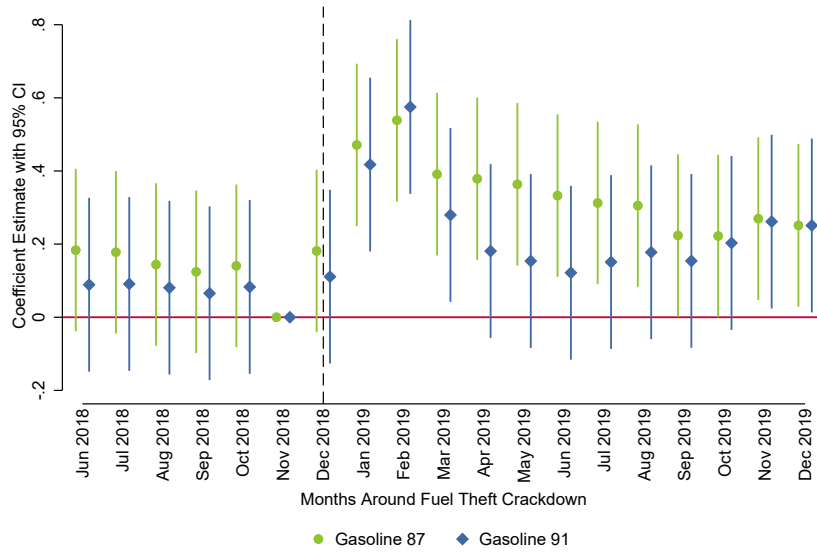
gasoline distribution. However, Pemex continues to own and operate most – though not all – production and transportation infrastructure (e.g., refineries and pipelines).

⁴Pipeline thefts also respond to fuel prices: we document a significant positive association between world oil prices and theft rates in Appendix 2.1.

In late December 2018, López Obrador declared a far-reaching crackdown on pipeline theft, which involved temporarily shutting down severely affected stretches of oil pipelines and using fuel trucks to transport refined oil products across those stretches, as well as increased military patrols and surveillance along oil pipelines (Crisis Group, 2022). This intervention resulted in widespread disruption of local fuel markets and fuel shortages (Specia, 2019).

In Figure 2, we use state-month level gasoline price data from Mexico’s Energy Regulatory Commission to compare gasoline prices in states where the Mexican government shut down pipelines between late December 2018 and April 2019 to combat fuel theft versus states where no pipeline shutdowns occurred (Caballero-Morales and Martínez-Flores, 2019; Carranza Garcia and Esposito, 2019). Estimates include month and state fixed effects. Results reveal a significant fuel price spike in states where the crackdown led to pipeline shutdowns (peaking at a price differential of up to 0.6 Mexican pesos per liter), beginning in January 2019 and persisting for at least a year.

Figure 2: Crackdown Effects on Gasoline Prices in Affected States



Note: Figure reports coefficient estimates from the following specification: $Y_{st} = \delta_s + \gamma_t + \sum_{t \neq 12/18} \beta_t T_{st} + \epsilon_{st}$, where Y_{st} are nominal gasoline 87 and 91 prices in Mexican Pesos per liter, δ_s and γ_t are state and month fixed effects, respectively, and T_{st} is an indicator taking a value of 1 in states where the Mexican government shut down oil pipelines in December 2018 to combat fuel theft. These states are identified as Hidalgo, México, Jalisco, Michoacán de Ocampo, Guanajuato, Querétaro, and Aguascalientes based on reports by Caballero-Morales and Martínez-Flores (2019) and Carranza Garcia and Esposito (2019). Data on fuel prices are from CRE (2023).

The most dramatic fallout from Mexico’s crackdown on fuel theft came on January 19th 2019, when a massive explosion resulting from a gasoline pipeline tap killed at least 74 people and injured hundreds more in the town of Tlahuelilpan, Hidalgo. Up to 800 people had converged on the gushing pipeline tap to collect fuel in jerrycans, in part because the crackdown had created local fuel shortages (Semple, 2019). Discussing the incident, residents explained that: “In these

towns, we all have a relative or friend who is dedicated to *huachicol* [pipeline theft]... Here, even the mayor protects *huachicol*... Authorities here receive money from *huachicol*. It pays very well and it's an opportunity to enjoy a better life (McDonnell and Linthicum, 2019)."

The high costs and substantial disruptions involved in the 2019 crackdown made it unsustainable in the long-term, leading to a winding down of crackdown efforts in April 2019. Since the crackdown did not resolve underlying security challenges, institutional weaknesses, and local incentives driving pipeline theft, illegal taps on refined oil pipelines began to grow again from 2021 onward (Crisis Group, 2022).

3 Theoretical Model

Section 2 suggests a clear pattern: successive law enforcement crackdowns on drug trafficking and oil theft inadvertently provoked cartel diversification into new illicit sectors. We formalize this mechanism in a model of sequential criminal adaptation under partial enforcement. We model a representative profit-maximizing cartel that allocates a limited stock of organizational capacity across drug trafficking, oil pipeline theft, and gas pipeline theft. By raising the relative cost of operating in the targeted sector, a crackdown prompts the cartel to reallocate capacity toward less-policed alternatives. This reallocation is facilitated by durable, sector-specific capital (e.g., technical expertise, corrupted officials, territorial networks) that is partially transferable to technologically and spatially adjacent sectors, reducing fixed costs of entry. Each new sector the cartel enters expands its organizational capacity, enabling it to sustain an ever-larger illicit portfolio. The result is a difficult-to-reverse *ratchet effect*: partial, sector-specific enforcement endogenously expands the criminal opportunity set.

Section 3.1 explains the model setup. Section 3.2 works through a sequential enforcement sequence corresponding to Mexico's crackdowns on drug trafficking and oil pipeline theft during the 2000s and 2010s, and derives propositions. Section 3.3 summarizes empirical predictions.

3.1 Setup

3.1.1 Sectors, Revenues, and Law Enforcement

Let $i \in \mathcal{I} = \{D, O, G\}$ denote three illicit sectors: drug trafficking, refined oil pipeline theft, and liquefied petroleum gas pipeline theft.⁵ The cartel can operate in any subset $A \subseteq \mathcal{I}$ simultaneously, choosing a scale of operation $s_i \geq 0$ for each active sector $i \in A$. Per-period

⁵In practice, cartels derive revenues from a wide range of illicit sectors. We focus on drug trafficking, oil theft, and gas theft to match our empirical setting. The model's insights generalize to any setting where criminal organizations face sector-specific enforcement and operate across activities with overlapping technological or spatial requirements, as discussed in Appendix 1.1.

profit from operating sector i at scale s_i is:

$$\Pi_i(s_i, e_i) = \pi_i(e_i) s_i - \frac{\gamma}{2} s_i^2, \quad (1)$$

where $\frac{\gamma}{2} s_i^2$ captures convex coordination costs.⁶ The term $\pi_i(e_i)$ is net revenue per unit of scale is $\pi_i(e_i) = R_i - C_i - \phi_i(e_i)$, where $R_i > 0$ is gross revenue per unit of scale, $C_i > 0$ is the baseline operating cost, and $\phi_i(e_i) \geq 0$ is the enforcement cost. The government exerts enforcement effort $e_i \geq 0$ targeted at sector i , subject to a capacity constraint $\sum_i e_i \leq E$ reflecting finite enforcement resources.⁷ Enforcement raises the expected cost of operating in sector i through increased probability of arrest and seizure, higher bribe payments required to maintain institutional protection under heightened scrutiny, and greater conflict with military and police forces. The enforcement cost function satisfies $\phi_i(0) = 0$ and $\phi'_i > 0$. We assume $\pi_i(0) > 0$ for all i , so that each sector is profitable absent enforcement.

3.1.2 Organizational Capacity

The cartel faces a limited stock of organizational capacity $M > 0$, the scarce managerial, financial, and logistical resources required to sustain criminal operations. Total scale across active sectors cannot exceed this capacity, making entry into a new sector costly even when it is independently profitable. Capacity itself grows with experience, as we formalize below.

3.1.3 Entry Costs and Sector-Specific Capital

The effective entry cost into sector i for a cartel holding capital K_j from sector j is:

$$\tilde{F}_i(K_j) = F_i [1 - \psi(\kappa_{ji}^\tau K_j^\tau, \kappa_{ji}^\sigma K_j^\sigma)], \quad (2)$$

where $F_i > 0$ is the greenfield entry cost of sector i , reflecting the fixed investment required to enter with no prior capital. Sector-specific capital K_i consists of: *technological capital* K_i^τ (e.g., sector-specific knowledge, techniques, equipment, and institutional capture of relevant officials) and *spatial capital* K_i^σ (e.g., territorial presence, control of physical corridors). For simplicity, each component is binary, $K_i^\tau, K_i^\sigma \in \{0, 1\}$, where 1 indicates that the cartel has operated in

⁶We adopt a quadratic cost specification for tractability, but the model's qualitative results hold for any strictly convex cost function. Convex coordination costs generate interior solutions in which the cartel partially withdraws from a cracked-down sector rather than exiting entirely. This is consistent with the empirical record: drug trafficking declined but persisted after the 2006 crackdown, and oil theft declined but persisted after 2019, with cartels adjusting scale across sectors simultaneously rather than fully exiting one upon entering another.

⁷Sector-specific enforcement is common in practice: governments respond to the salience of harms associated with specific illicit activities. In contrast, general anti-crime efforts (i.e., institutional or anti-corruption measures) can be politically costly, difficult to implement, and less visible to the public. For simplicity, we assume the government is myopic, responding to criminal activity in the targeted sector without anticipating diversification.

sector i and acquired the relevant capital. One stage of operation suffices to acquire the full stock, which is durable: connections, expertise, and territorial presence persist into subsequent stages regardless of scale reductions.

The parameters $\kappa_{ij}^\tau \in [0, 1]$ and $\kappa_{ij}^\sigma \in [0, 1]$ denote the *technological* and *spatial transferability* between the sectors i and j , with higher values indicating greater similarity. They measure the extent to which capital acquired in one sector can be redeployed in another. The proportional reduction in entry costs due to previously accumulated technological and spatial capital is captured by $\psi(\kappa_{ji}^\tau K_j^\tau, \kappa_{ji}^\sigma K_j^\sigma)$, which is increasing in both arguments with $\psi(0, 0) = 0$. When the cartel holds capital from multiple prior sectors, the relevant source is the one with the highest total transferability. We assume $\tilde{F}_i$ is positive in all cases.

Since operating in a new sector expands the cartel's stock of sector-specific capital, it also expands organizational capacity: $M = M(K^-)$ is increasing in previously accumulated capital, with $\partial M / \partial K_i^{-\tau} > 0$ and $\partial M / \partial K_i^{-\sigma} > 0$ for all i . We denote by $\Delta M_i > 0$ the resulting increase in capacity following one full stage of operation in sector i .

3.1.4 The Cartel's Portfolio Problem

In each stage, the cartel chooses a portfolio $A \subseteq \mathcal{I}$ and scales of operation $\{s_i\}_{i \in A}$ to maximize total profit net of entry costs into new sectors:

$$\max_{A, \{s_i\}} \sum_{i \in A} \Pi_i(s_i, e_i) - \sum_{i \in A \setminus A^-} \tilde{F}_i(K^-) \quad \text{subject to} \quad \sum_{i \in A} s_i \leq M, \quad (3)$$

where A^- is the portfolio from the previous stage.⁸ The first-order condition for each active sector yields:

$$s_i^* = \frac{\pi_i(e_i) - \lambda}{\gamma}, \quad (4)$$

where $\lambda \geq 0$, the shadow price of capacity, is positive when the constraint binds and zero otherwise. A crackdown on sector i reduces $\pi_i(e_i)$ and thus s_i^* , but the sector remains active at reduced scale as long as $\pi_i(e_i) > \lambda$. The resulting fall in s_i^* also lowers λ , reducing the opportunity cost of deploying capacity in a new sector. Entry into sector j is worthwhile only if the gain from optimally reallocating M across $A \cup \{j\}$ exceeds the effective entry cost $\tilde{F}_j(K^-)$.

⁸We assume the cartel solves this problem myopically, taking current enforcement as given and not anticipating future strengthening or weakening of enforcement.

3.2 Sequential Enforcement and the Ratchet Effect

In this section, we analyze how the solution to equation (3) evolves across a sequence of sector-specific crackdowns. We focus on the Mexican context of drug trafficking and oil and gas theft, and introduce an assumption reflecting empirically observed conditions in this setting, which we then use to show how the model predicts the observed patterns of cartel diversification following the War on Drugs and the 2019 oil theft crackdown.

Assumption 1 (Transferability Between Drugs, Oil Theft, & Gas Theft). *Technological transferability satisfies $\kappa_{OG}^\tau > \kappa_{DO}^\tau > \kappa_{DG}^\tau$, and spatial transferability satisfies $\kappa_{OG}^\sigma > \kappa_{DO}^\sigma \approx \kappa_{DG}^\sigma$.*

Oil and gas theft share overlapping institutional features (e.g., contact with pipeline operators, technical methods), and partially overlapping geographies, making capital used in oil theft highly transferable to gas operations (high $\kappa_{OG}^\tau, \kappa_{OG}^\sigma$). By contrast, drug trafficking relies on distinct networks, logistics, and territorial control, implying low transferability to either pipeline sectors ($\kappa_{DO}^\tau, \kappa_{DG}^\tau, \kappa_{DO}^\sigma, \kappa_{DG}^\sigma$). Within this range, technological transferability is higher from drugs to oil than to gas ($\kappa_{DO}^\tau > \kappa_{DG}^\tau$), reflecting the greater technical complexity of gas theft. Transferability coefficients for the three sectors are summarized in Appendix 1.1, which also discusses how the same framework can reproduce diversification patterns in other illicit sectors by imposing alternative transferability structures.

3.2.1 Stage 0: Pre-Crackdown Baseline

At baseline, enforcement $e_i^0 \approx 0$ for all i . The cartel operates in D alone with capacity M_0 and drug capital $K_D = (1, 1)$. We assume λ_0 is high enough that entry into oil or gas is not profitable, consistent with near-zero pipeline theft prior to Mexico's War on Drugs in 2006.⁹

3.2.2 Stage 1: Crackdown on Drug Trafficking

In this stage, the government raises enforcement against drug trafficking to $e_D^1 \gg e_D^0$, reducing $\pi_D(e_D^1)$. With only D active, the decline in the cartel's profitability lowers its opportunity cost of diverting capacity to a new sector, such that $\lambda_1 < \lambda_0$. As derived in Appendix 1.2, a cartel with drug capital $K_D = (1, 1)$ enters sector $j \in \{O, G\}$ if and only if the profit gain from optimally reallocating M_0 across $\{D, j\}$ exceeds the effective entry cost:

$$\pi_j(e_j^1) > \lambda_1 + 2\sqrt{\gamma \tilde{F}_j(K_D)} \equiv \bar{\pi}_j. \quad (5)$$

⁹The capacity constraint binds at shadow price $\lambda_0 = \pi_D(e_D^0) - \gamma M_0 > 0$. This holds when M_0 is sufficiently small in relation to drug profitability, making the opportunity cost of diversification prohibitive.

Proposition 1 (Sequential Entry in Stage 1). *Under Assumption 1, the entry threshold for gas strictly exceeds that for oil ($\bar{\pi}_G - \bar{\pi}_O > 0$). There therefore exists a non-empty range of oil profits $\pi_O(e_O^1) \in (\bar{\pi}_O, \bar{\pi}_G]$ under which the drug crackdown induces oil entry but not gas entry (proof in Appendix 1.3).*

We proceed under the empirically relevant configuration in which $\pi_O(e_O^1) \in (\bar{\pi}_O, \bar{\pi}_G]$. Drug operations scale back to $s_D^* < M_0$ to accommodate oil but do not cease entirely, in line with empirical patterns. Having operated O for one full period, the cartel accumulates $K_O = (1, 1)$ and capacity grows to $M_1 = M_0 + \Delta M_O$.

3.2.3 Stage 2: Crackdown on Oil Pipeline Theft

In this stage (corresponding with the 2019 crackdown), the government raises enforcement against oil theft ($e_O^2 \gg e_O^1$) while maintaining drug enforcement at e_D^1 . The cartel enters Stage 2 with drug capital $K_D = (1, 1)$, oil capital $K_O = (1, 1)$, and expanded capacity $M_1 > M_0$.¹⁰ With both D and O active under binding constraint M_1 , the Stage 2 shadow price is:

$$\lambda_2 = \frac{\pi_D(e_D^1) + \pi_O(e_O^2)}{2} - \frac{\gamma M_1}{2}. \quad (6)$$

Both the oil crackdown (reducing π_O) and expanded capacity ($M_1 > M_0$) put downward pressure on λ_2 , lowering the opportunity cost of adding gas. The entry cost into gas is determined by $K_O = (1, 1)$, the most applicable capital stock under Assumption 1. As derived in Appendix 1.2, the Stage 2 entry threshold for gas is:

$$\pi_G(e_G^2) > \lambda_2 + \sqrt{3\gamma \tilde{F}_G(K_O)} \equiv \bar{\pi}_G^2. \quad (7)$$

Two distinct forces lower $\bar{\pi}_G^2$ relative to the Stage 1 gas threshold $\bar{\pi}_G^1$, each operating through a different term. The *capital channel* lowers $\tilde{F}_G$: oil experience provides a larger entry subsidy than drug experience. The *capacity channel* lowers λ_2 : both the oil crackdown and expanded M_1 reduce the opportunity cost of adding gas. Whether $\bar{\pi}_G^2 < \bar{\pi}_G^1$ holds depends on parameter magnitudes. Having entered gas in Stage 2 and operated for one full period, the cartel accumulates $K_G = (1, 1)$ and capacity grows to $M_2 = M_1 + \Delta M_G$.

The enforcement sequence generates propositions that formalize the model's mechanisms.

Proposition 2 (Partial Displacement). *In a multi-sector portfolio, a crackdown on sector i reduces $s_i^* = (\pi_i(e_i) - \lambda)/\gamma$ and reallocates capacity toward other active sectors, as long as*

¹⁰We assume the oil theft crackdown is not so severe as to eliminate oil theft entirely, i.e., $\pi_O(e_O^2) > \pi_D(e_D^1) - \gamma M_1$, so oil remains active at reduced scale, in line with empirical patterns.

$$\pi_i(e_i) > \lambda.$$

The convex coordination costs generate this interior solution: the cartel optimally withdraws partially from a cracked-down sector rather than exiting entirely. Accumulated capital K_i persists regardless of the scale reduction.

Proposition 3 (Ratchet Effect). *Having traversed $D \rightarrow O$, the cartel faces a lower gas entry threshold in Stage 2 than it would without oil experience, through two channels:*

(i) **Capital channel.** *Since $\kappa_{OG}^\tau > \kappa_{DG}^\tau$ and $\kappa_{OG}^\sigma > \kappa_{DG}^\sigma$ under Assumption 1, oil experience generates a strictly larger proportional reduction in F_G than drug experience, so $\tilde{F}_G(K_O) < \tilde{F}_G(K_D)$. The capital subsidy $F_G[\psi(\kappa_{OG}^\tau, \kappa_{OG}^\sigma) - \psi(\kappa_{DG}^\tau, \kappa_{DG}^\sigma)] > 0$ decomposes into a technological component ($\kappa_{OG}^\tau > \kappa_{DG}^\tau$), given that oil expertise lowers the cost of gas operations, and a spatial component ($\kappa_{OG}^\sigma > \kappa_{DG}^\sigma$), given that territorial presence near oil infrastructure lowers the cost of entering the overlapping gas network.*

(ii) **Capacity channel.** *The expanded capacity $M_1 = M_0 + \Delta M_O$ reduces λ_2 relative to the counterfactual without oil experience, lowering the opportunity cost of adding gas.*

Proposition 4 (Opportunity Set Expansion). *Gas theft was blocked in Stage 1 because $\pi_G(e_G^1) \leq \bar{\pi}_G^1$ (Proposition 1). Through the ratchet effect, both channels of Proposition 3 lower $\bar{\pi}_G^2$ relative to $\bar{\pi}_G^1$. There therefore exist parameter values under which $\bar{\pi}_G^2 < \bar{\pi}_G^1$ and gas entry becomes profitable in Stage 2 even though $G \notin A^0$ (see Appendix 1.4 for the sufficient condition). Sequential crackdowns can endogenously enable criminal opportunities that did not exist at baseline.*

3.2.4 Stage 3: Relaxation of Oil Theft Crackdown

Now suppose the crackdown on oil theft eases ($e_O^3 < e_O^2$).

Proposition 5 (Persistence of Diversification). *Following a relaxation of enforcement against oil theft, the cartel scales up oil operations. Gas operations nonetheless persist at positive scale as long as $\pi_G(e_G) > [\pi_D(e_D^1) + \pi_O(e_O^3) - \gamma M_2]/2$, which is decreasing in M_2 : the larger the capacity accumulated through diversification, the easier it is for all three sectors to coexist.*

Two mechanisms sustain gas theft when oil enforcement eases. First, gas capital $K_G = (1, 1)$ is durable: connections, knowledge, and territorial presence built in Stage 2 do not depreciate when oil enforcement relaxes. Second, the entry cost $\tilde{F}_G(K_O)$ is sunk, so the cartel faces only a scale decision in Stage 3 rather than a re-entry decision. Gas theft persists as long as its net revenue exceeds the shadow price of capacity, which is kept low by expanded capacity $M_2 > M_0$.

Corollary 1 (Irreversibility of Diversification). *No enforcement policy that operates solely through $\phi_i(e_i)$ can restore the Stage 0 criminal opportunity set. Gas capital $K_G = (1, 1)$ is durable, the entry cost $\tilde{F}_G(K_O)$ is sunk, and capacity $M_2 > M_0$ is permanent. Sequential crack-downs have irreversibly enlarged the cartel’s portfolio.*¹¹

3.2.5 Scope of the Model

Beyond Assumption 1, three features of the model are assumed rather than derived from the cartel’s optimization problem: (i) sector-specific capital is fully acquired within a single stage of operation, (ii) this capital does not subsequently depreciate, and (iii) organizational capacity expands permanently with each new sector entered.¹² These primitives do not bear equally on every proposition. Proposition 2 is a derived result, following from the interior-solution of the cartel’s optimization problem. Propositions 3 and 4 additionally require that capital and capacity gains from entry into a new sector survive into the immediately subsequent stage – a comparatively weak form of durability. Proposition 5 and Corollary 1 require the stronger claim that these gains persist indefinitely. We find empirical support for this stronger form of durability in observed persistence and bounce-back of oil theft during and after the crackdown on that sector. However, we stress that this is an assumption rather than a derived result. In Appendix 1.6, we discuss how weaker durability assumptions affect each model proposition.

3.3 Empirical Predictions

The propositions above generate predictions that guide our empirical analysis. We do not test predictions related to the first-stage diversification from drug trafficking into oil theft, as this transition is well-documented in existing work (Battiston et al., 2024; López Cruz and Torrens, 2023; Dell, 2015). Instead, we focus on the previously unstudied second-stage diversification from oil theft into gas theft.

1. Gas theft increases after the crackdown on oil theft. Propositions 3 and 4 predict

¹¹Reversing the ratchet effect requires either (i) degrading sector-specific capital K_i and thus reducing organizational capacity M directly (for instance, by prosecuting corrupted officials or targeting cartels’ financial operations), or (ii) spreading enforcement across sectors simultaneously to prevent the shadow price of capacity from falling far enough to make diversification worthwhile. We characterize the optimal multi-sector enforcement allocation in Appendix 1.5.

¹²These assumptions are based on observation of the historical record and what we believe are reasonable simplifications. The assumption of single-period capital accumulation reflects the model’s discrete-stage structure. For the drugs-to-oil-to-gas sequence we study, each stage spans roughly five to ten years in real time, which is plausibly long enough for cartels to accumulate relevant skills, territory, and connections. The assumption that sector-specific capital does not depreciate is motivated by the empirical record: cartels that come under increased enforcement in a sector scale down activity within that sector (i.e., adjust the intensive margin, s_i) but do not fully exit it, consistent with accumulated knowledge, territorial presence, and institutional connections (i.e., the extensive margin) persisting even as scale declines. The assumption of permanent capacity expansion upon entry into a new sector follows directly from non-depreciation, since capacity is a function of accumulated capital.

that the oil theft crackdown lowers the gas entry threshold $\bar{\pi}_G^2$ below its Stage 1 value through the capital and capacity channels, making gas entry profitable in Stage 2 even though it was not profitable at any prior stage.

2. **Spatial channel: gas theft increases most in municipalities that neighbor or host oil pipelines.** The cartel’s existing territorial presence near oil infrastructure lowers the fixed costs of establishing gas theft operations in these municipalities.
3. **Technological channel: diversification into gas theft is strongest among cartels with prior experience in oil theft.** These cartels possess institutional and technical knowledge directly applicable to gas theft operations, lowering their barriers to entry relative to cartels without oil experience.
4. **Persistence of gas theft after the oil theft crackdown ends.** Proposition 5 and Corollary 1 predict that when oil enforcement eases, oil theft will recover while gas theft will not contract. Gas capital is durable, the entry cost is sunk, and expanded capacity M_2 sustains simultaneous activity in both sectors.

Cartel diversification into gas theft is also likely to increase violence, both through territorial expansion that triggers turf wars and through violence used to coerce officials and enforce control in newly entered territories (Sobrinho, 2020; Atuesta and Ponce, 2017; Badillo-Sarmiento and Trejos-Rosero, 2024). This generates a fifth empirical prediction:

5. **Homicides increase in municipalities hosting gas pipelines following the oil theft crackdown.** Based on the above predictions, violence should be concentrated in municipalities near oil pipelines and that experienced cartel growth.

4 Data

Our sample consists of 2,471 municipalities tracked annually between 2015-2022. In this section, we describe each of our data sources and present descriptive statistics.

4.1 Energy Infrastructure

We obtained georeferenced information on Mexico’s energy infrastructure from the National Commission of Hydrocarbons (Comisión Nacional de Hidrocarburos) (CNH, 2019). Data include precise locations of oil refineries, gas processing complexes, and gas compression centers, as well as pipelines used in the transportation of crude, refined oil, and gas products, as of 2019. Among

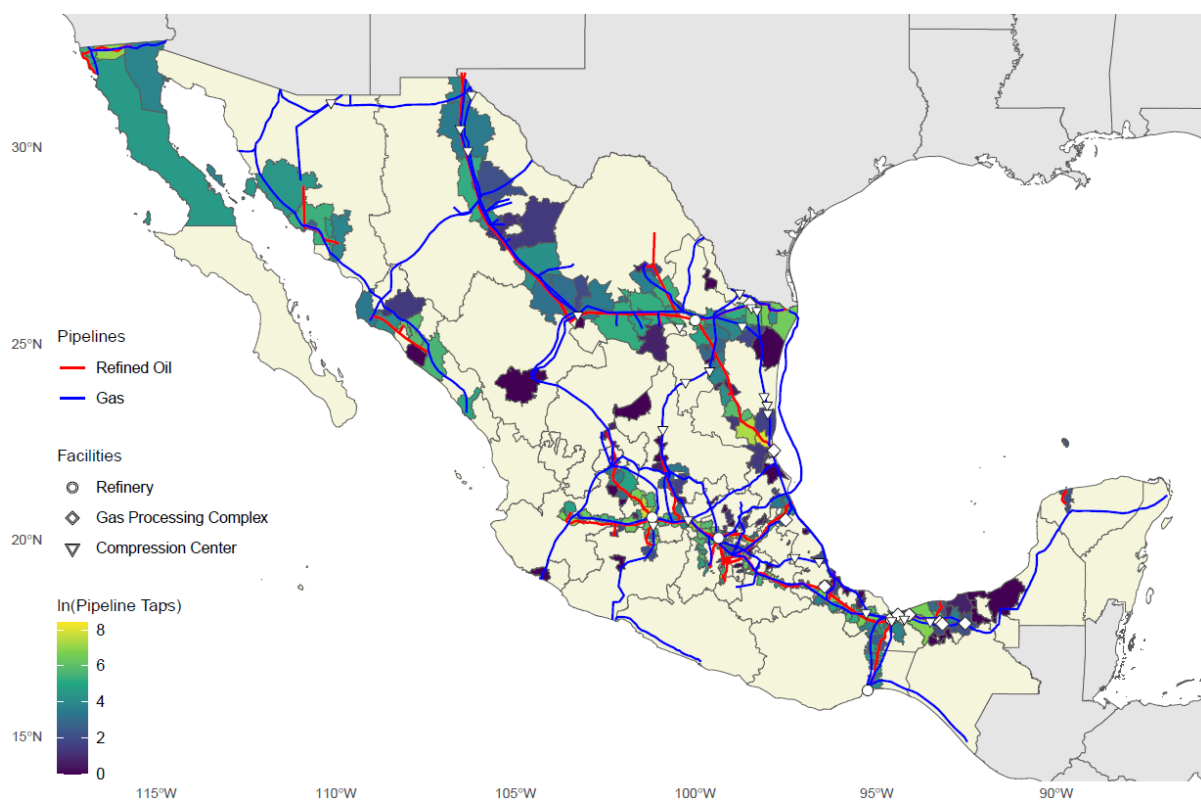
oil pipelines, we focus on those designated as carrying gasoline and diesel – the products targeted by fuel thieves – and exclude pipelines that exclusively carry other products, such as crude oil, fuel oil (a heavy fuel used by industry), and jet fuel.

Mexico’s gas distribution infrastructure includes government-owned pipelines (such as the “SISTRANGAS” system) and privately owned pipelines. We retain both in our main specifications. We exclude importation pipelines from our analysis, as these are located outside the Mexican territory. Figure 3 maps Mexico’s refined oil and gas pipelines, as well as the locations of refineries, gas processing complexes, and gas compression centers.

4.2 Illegal Taps on Oil and Gas Pipelines

Data on reports of illegal gasoline, diesel, and LPG pipeline taps were obtained through a freedom of information request to Pemex. The data include the number of illegal taps reported each month in each Mexican municipality from 2010 to 2022. Taps on refined fuel and gas products are reported separately. In Figure 3, we overlay cumulative oil and gas pipeline taps reported in each municipality between 2010-2022.

Figure 3: Mexico: Refined Oil and Gas Processing and Transportation Infrastructure



Note: Geo-referenced data on pipelines and other energy infrastructure are from the National Commission of Hydrocarbons (CNH, 2019); data on pipeline taps were obtained from Pemex through freedom of information requests. Gas pipelines, plotted in blue, include both government- and privately owned infrastructure.

4.3 Crime Reports

We use data on homicides from the National Department of Health Information (Sistema Nacional de Información en Salud – SINAIS). This database includes information on the cause of death, location of death, and type of weapon used in all homicides. It also reports the sex, occupation, marital status, and insurance affiliation of the deceased.¹³

Using municipal data on population from the National Institute of Statistics and Geography (INEGI), we calculate homicide rates per 100,000 people for males, females, potentially cartel-related individuals, Pemex employees, and military personnel. Potentially cartel-related homicides are defined as deaths of men aged 15-40 who were killed by firearms (De Haro, 2025). We use this definition because homicides of men within this age group have been shown to closely match the geographic and temporal patterns of reported executions associated with cartel rivalry in Mexico (Calderón et al., 2015). Furthermore, civilian access to firearms in the country is highly restricted, and over 70% of all guns recovered at crime scenes can be traced back to drug cartel organizations (Mineo, 2022).

As a measure of deaths from combat between the military and cartels, we identify homicides of individuals whose insurance affiliation was with SEDENA, Mexico’s Defense Ministry. Since military family members can also have this affiliation, we limit this definition to homicides of working-age SEDENA-affiliates (18-60 years old) killed by firearms. Similarly, we identify homicides of Pemex employees as deaths of working-age Pemex-affiliates by firearms. Data on other crimes were obtained from the Ministry of Public Security (SSP). This database includes information on all crimes reported to the police in each municipality. We focus on robberies, kidnappings, threats and extortion, and non-lethal physical violence.

4.4 Cartel Presence

The Mapping Criminal Organizations project (Signoret et al., 2017) provides a municipality-year panel dataset on the presence of criminal organizations in Mexico. The project employs a web-crawling technique to identify news articles related to criminal cartels on Google and Google News, drawing from over 80 international, 30 national, and 750 local news outlets (Sobrinho, 2020). A cartel is recorded as present in a municipality if it is mentioned at least once in a given year.¹⁴ The database contains information on the presence of 75 criminal organizations

¹³We prefer this database to police report data from the Ministry of Public Security (SSP) because it provides more details on the victim, including insurance affiliation, which we use to identify murders of Pemex employees and military personnel. It also does not suffer from under-reporting of homicides, as police reports might, since it is based on death certificates rather than police investigations.

¹⁴The MCO dataset relies on news reports, which could be biased if criminal organizations affect news reporting. This potential bias is minimized by the dataset’s design and coverage. Since a cartel is recorded as present if mentioned at least once in a year, over-reporting in municipalities with more media attention does not inflate

in Mexico at the municipal level from 1990 to 2021. Appendix Table A3 reports baseline, pre-crackdown (2015-2018) summary statistics for (i) municipalities with either a gas or refined oil pipeline and (ii) non-pipeline municipalities.¹⁵

5 Empirical Strategy

We implement a difference-in-differences (DiD) strategy to estimate how the government’s fuel theft crackdown impacted illicit pipeline tapping, homicide rates, cartel presence, and other crime outcomes in directly affected municipalities (i.e., those with oil pipelines) and indirectly affected municipalities (i.e., those with gas pipelines), relative to control municipalities that do not host pipelines.¹⁶ Our main specification is as follows:

$$Y_{it} = \alpha_i + \tau_t + \delta(P_i^G \times Post_{t \geq 2019}) + \gamma(P_i^O \times Post_{t \geq 2019}) + e_{it} \quad (8)$$

where Y_{it} is an outcome of interest in municipality i and year t and P_i^G and P_i^O are binary variables indicating the presence of pipelines carrying gas and refined oil products, respectively. $Post_{t \geq 2019}$ indicates the period before and after the military crackdown on fuel theft in 2019. Finally, we control for time-invariant municipality characteristics and temporal shocks using municipality (α_i) and year-fixed effects (τ_t). Municipalities may host both oil and gas pipelines, and this specification flexibly allows P_i^G and/or P_i^O to equal 1.

Our primary parameter of interest in this specification is δ , which measures change in the outcome in municipalities with a gas pipeline after the crackdown, relative to municipalities that do not host gas pipelines. If criminal organizations diversified into gas pipeline theft, we expect to observe an increase in homicide rates and cartel presence in municipalities with a gas pipeline ($\hat{\delta} > 0$). A secondary parameter of interest is γ , which measures the direct effect of

measures of presence. Under-reporting is mitigated through inclusion of a broad set of national and international media outlets, which are less likely to be suppressed by local intimidation (Sobrinho, 2020).

¹⁵We collect supplementary data on socioeconomic characteristics of municipalities from the National Association of Universities and Higher Education Institutions (Asociación Nacional de Universidades e Instituciones de Educación Superior – ANUIES) and the Telephone Survey of Occupation and Employment (Encuesta Telefónica de Ocupación y Empleo – ETOE). Municipalities hosting a pipeline exhibit higher homicide, cartel presence, and crime rates than the rest of the country. Socioeconomic indicators are similar for places with and without pipelines. Pipeline-hosting municipalities have larger populations, and we report main results as rates per 100k residents to account for these differences.

¹⁶Ideally, we would be able to identify the exact location and date of crackdown operations to achieve a more precise definition of municipalities’ treatment status. However, repeated freedom of information requests to Mexico’s Federal Police, Ministry of Defense, and Pemex were unsuccessful due to the sensitive national security nature of these data. Furthermore, while specific states were targeted with pipeline shutdowns during the 2019 crackdown, military and police anti-theft operations extended broadly across areas hosting refined oil pipelines. Data on military operations against pipeline thefts that we were able to obtain for 2021 indicate operations occurring in at least 17 states in that year. Thus, we consider all municipalities hosting refined oil pipelines to be directly treated by the crackdown, and all municipalities hosting gas pipelines to be indirectly treated by potential cartel substitution toward gas theft.

the crackdown on places with oil pipelines. For outcomes such as homicide rates and cartel presence, negative values of $\hat{\gamma}$ would suggest the crackdown was associated with a reduction of crime in directly targeted oil-pipeline municipalities. In our preferred specification, we normalize homicide and crime rates to number of incidents per 100,000 municipal inhabitants to account for large differences in population size between municipalities.

We also estimate event studies to assess dynamic effects of the crackdown:

$$Y_{it} = \alpha_i + \tau_t + \sum_{k \neq 2018} \delta_k P_i^G \times I(t = k) + \sum_{k \neq 2018} \gamma_k P_i^O \times I(t = k) + v_{it} \quad (9)$$

In this specification, coefficients δ_k and γ_k capture the impact of the crackdown k years before or after 2019 on outcome Y_{it} in municipalities that host gas and refined oil pipelines, respectively. Pre-crackdown year 2018 is omitted as a baseline reference. Besides revealing dynamic treatment effects across years, event studies also allow evaluation of the identifying parallel pre-trends assumption that underlies DiD strategies.¹⁷

5.1 Identification

There are several potential threats to identification in our empirical strategy. First, reverse causality could occur if direct treatment (i.e., directly experiencing the military crackdown due to the presence of oil pipelines) was caused by trends in outcomes of interest during pre-treatment periods. As shown in the results below, this is indeed the case for oil pipeline municipalities: escalating rates of oil theft led the government to focus its military crackdown on those areas. While oil pipeline areas were thus *endogenously* treated and exhibit significant pre-trends in theft outcomes, gas pipeline municipalities were *exogenously* treated by spillovers from a crackdown focused elsewhere.

Although some municipalities host both oil and gas pipelines, crackdown targeting was based solely on the location of high-theft oil pipelines, a criterion that is plausibly exogenous to whether a municipality also hosts gas infrastructure. Furthermore, by including both $P_i^O \times Post$ and $P_i^G \times Post$ simultaneously in Equation 8, the coefficient δ on gas pipeline presence is estimated conditional on oil pipeline presence. Any post-crackdown changes in overlap municipalities attributable to direct targeting are thus absorbed by γ , reducing the concern that δ reflects

¹⁷For theft outcomes (i.e., illicit taps on oil and gas pipelines) we estimate Equation 9 separately by fuel type, including only the $P_i^f \times I(t = k)$ interactions for the relevant fuel type $f \in \{\text{gas, refined oil}\}$ rather than both jointly. Theft counts are mechanically zero in municipalities without pipelines of type f , so municipalities hosting only the other pipeline type provide no meaningful comparison for theft of type f . Identification therefore comes from within-municipality variation over time in pipeline municipalities, rather than from cross-unit comparison to a control group.

direct enforcement effects rather than spillovers driven by cartel substitution toward gas theft. We exploit heterogeneity across overlap and non-overlap gas pipeline municipalities in Section 6 to further probe spatial patterns of diversification and violence. Throughout Section 6, we interpret results for oil pipeline municipalities as descriptive. For gas pipeline municipalities, we interpret pipeline theft results as causal because these places were exposed to a plausibly exogenous shock. Homicides and cartel presence, by contrast, are downstream equilibrium outcomes of theft substitution that may also reflect contemporaneous territorial conflict. We thus interpret these results as plausibly, though not definitively, causal.

A second threat to identification could come from violations of the stable unit treatment value assumption (SUTVA) if spillovers from treated groups contaminate the control group (i.e., non-pipeline municipalities). We quantify spatial spillovers explicitly in Section 6.5. We also estimate a “spillover-free” robustness check in Section 7 that excludes control municipalities sharing a border with places hosting oil or gas pipelines and find similar results.

Third, territorial reorganization by cartels unrelated to the oil theft crackdown could confound our estimates. Specifically, López Cruz and Torrens (2026) document a fentanyl-driven wave of territorial expansion and route re-optimization by the *Cártel de Jalisco Nueva Generación* (CJNG) beginning in 2013, which overlaps geographically with pipeline theft areas. Moreover, from 2018 onward, the Mexican government adopted a less aggressive posture toward organized crime under the “*abrazos no balazos*” security strategy, potentially enabling cartel expansion independently of the crackdown. We address this concern in three ways. First, because fentanyl-driven reorganization (starting in 2013) largely predates our study period, its effects on CJNG’s territorial footprint should be captured in pre-2019 cartel presence. As a robustness check, we control for 2015-2018 CJNG presence interacted with year fixed effects, thereby absorbing differential pre-treatment trends in municipalities where CJNG was already established. Second, we verify that increased cartel presence in gas-pipeline municipalities after the crackdown was not driven entirely by CJNG. Third, we re-estimate event studies omitting municipalities where fentanyl seizures occurred during the study period. These checks, however, cannot rule out entirely that CJNG’s fentanyl-driven territorial reorganization shaped where displaced groups chose to relocate in response to the crackdown and why violence rose in those areas, making it a complementary confounder.

Finally, our estimates could suffer from omitted variable bias if treated municipalities (those with oil or gas pipelines) and control municipalities differ in ways that systematically affect pre- and post-treatment trends. In our preferred specifications, we include municipality and year fixed effects to reduce these concerns. In Section 7, we estimate a robustness check that

limits the sample to treated and control municipalities that exactly match on binned baseline characteristics – including matching weights – ensuring that treated and control groups are comparable on a rich set of baseline observables. We find that results are strongly robust. Results are also robust to controlling for municipal Covid-19 rates, reducing concerns over another potential confounder. Finally, we note that nearly all pipelines were constructed prior to 2018, and most gas pipelines were constructed in the 1940s-80s – with their location determined by engineering considerations and cost-minimizing routes. Siting of pipelines was thus orthogonal to socioeconomic and crime characteristics of municipalities decades later.

6 Results

This section presents empirical tests of the predictions outlined in Section 3. We begin by documenting the first-order effect of the crackdown: an increase in illegal taps on gas pipelines (Prediction 1). We then examine the spatial distribution of this increase, showing that it concentrates in municipalities near or hosting oil pipelines (Prediction 2). Next, we turn to cartel presence, showing that cartels expand into gas-pipeline municipalities following the crackdown, with the largest effect among cartels with prior oil theft experience (Prediction 3). The event studies documenting Predictions 1 and 2 also reveal that gas theft persists after the crackdown on oil theft eases (Prediction 4). Finally, we document increased homicide rates in gas-pipeline municipalities after the crackdown, with effects concentrated in areas near oil pipelines that experienced cartel growth (Prediction 5). Finally, we briefly discuss crackdown effects on other crimes and spatial spillovers onto neighboring municipalities.

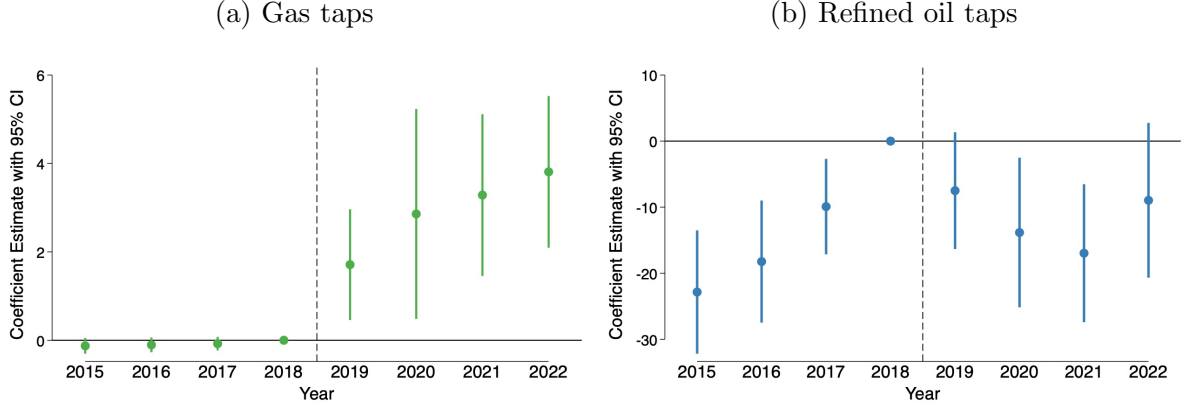
As noted in Section 5 above, oil pipeline municipalities were endogenously targeted due to upward-trending levels of theft, leading us to interpret results for these places descriptively. In contrast, gas pipeline municipalities were affected indirectly by spillovers from the crackdown on oil pipeline municipalities, making their treatment plausibly exogenous.

6.1 Refined Oil and Gas Theft

Figure 4 reports estimated effects of the fuel theft crackdown on (a) illegal taps on gas pipelines and (b) illegal taps on refined oil pipelines. As shown in Panel (a), there is no trend in gas pipeline thefts prior to the 2019 crackdown, after which gas thefts increase significantly, rising by an average of 1.7 thefts per municipality-year in 2019 and by 3.8 thefts per municipality-year by 2022. As shown in Panel (b), illegal taps on refined oil pipelines were trending sharply upwards in years preceding the military crackdown, which is precisely what triggered the crackdown in the first place. The strong pre-trend in this panel is thus to be expected. The 2019 crackdown

was associated with a temporary reversal in this upward trend and significant reduction in oil thefts for several years (relative to their 2018 level), before thefts begin trending back toward previous levels in 2022.

Figure 4: Effects of the Fuel Theft Crackdown on Illegal Pipeline Taps



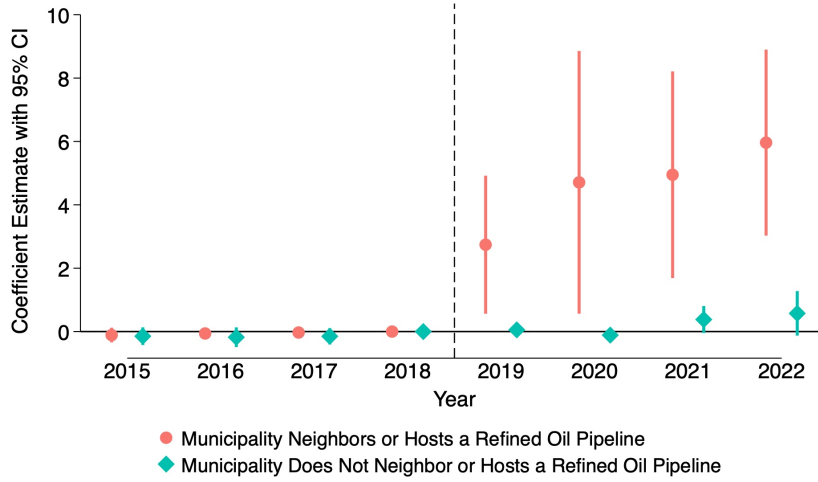
Note: Figure reports coefficient estimates and 95% confidence intervals for (a) number of illegal taps reported on gas pipelines (liquid petroleum gas and natural gas), and (b) number of illegal taps reported on refined oil (gasoline and diesel) pipelines. This specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

We next estimate separate gas theft event studies for municipalities that neighbor or host a refined oil pipeline and those that do not. Figure 5 shows that the post-crackdown increase in gas pipeline thefts is concentrated entirely in municipalities that neighbor or host a refined oil pipeline, while municipalities with no proximity to oil infrastructure show no discernible increase.¹⁸ Together, the results in Figures 4 and 5 support Predictions 1 and 2: the crackdown triggered a measurable shift into gas theft, and this shift was geographically anchored by cartels' existing presence along oil pipeline corridors, consistent with spatial capital transferability lowering fixed costs of entry into gas theft.

Appendix Figure A3 further disaggregates this result into municipalities that (i) host an oil pipeline, (i) neighbor a municipality with an oil pipeline, or (iii) neither host nor neighbor an oil pipeline. Effects of the crackdown on gas theft are strongest inside oil-hosting municipalities themselves. This pattern indicates highly localized substitution: the 2019 crackdown targeted oil pipelines and transport specifically, with enforcement tightly focused on oil infrastructure rather than gas pipelines. As Appendix Figure A4 illustrates, oil and gas pipelines often run

¹⁸Based on media reporting that gas pipeline theft is easier in places where gas pressure is higher (Nájjar, 2019), we compute the distance between municipality centroids and the nearest gas compression center (facilities that maintain gas pressure along pipelines). We then re-estimate event studies of gas pipeline thefts for places near (≤ 50 km) and far (> 50 km) from a compression center. Specifically, we estimate a four-way specification to explore joint effects of neighboring an oil pipeline (where spatial substitution from oil to gas theft is easier) and neighboring a compression center. Results, reported in Appendix Figure A2, indicate that post-crackdown increases in gas taps are driven by proximity to oil pipelines, with no significant difference based on proximity to a compression center. This suggests that any advantage in gas theft resulting from optimal pipeline pressure is not large enough to overcome the fixed costs of entering distant new territories.

Figure 5: Crackdown Effects on Gas Thefts Near/Far from Oil Pipelines



Note: Figure reports coefficient estimates and 95% confidence intervals for the number of illegal gas taps reported in municipalities hosting gas pipelines, decomposed into two groups: (i) municipalities that neighbor or host a refined oil pipeline, and (ii) municipalities that do not neighbor or host a refined oil pipeline. Specification includes municipality and year fixed effects, and standard errors are clustered at the municipality level.

through distinct parts of the same municipality, meaning cartels could shift tapping activity to gas pipelines within their existing territorial footprint without running afoul of active oil theft enforcement. This localized pattern of spatial substitution is consistent with Prediction 2, i.e., that diversification into gas theft will concentrate where existing territorial networks lower the fixed costs of entry.

These findings also support Prediction 4 on the persistence and irreversibility of diversification. Although the pipeline shutdown ended four months after the 2009 crackdown began, Figures 1 and 5 show that gas theft continued to increase well after enforcement pressure subsided. In contrast, oil theft declined during the crackdown but only gradually returned toward pre-crackdown levels thereafter. In other words, cartels did not revert to oil theft once enforcement relaxed. They remained active in gas theft and simultaneously rebuilt oil theft operations, consistent with the ratchet mechanism in the model, whereby entry into a new sector expands organizational capacity and generates durable capital.

6.2 Cartel Presence

Table 1 and Figure 6 report estimated crackdown effects on cartel presence, which we use to test Prediction 3: that diversification into gas theft is driven disproportionately by cartels with prior experience in oil theft operations.

We first establish that cartel presence in gas pipeline municipalities increased following the crackdown. Columns (1) and (4) of Table 1 report effects on the number of cartels (intensive margin) and on a binary indicator of any cartel presence (extensive margin), respectively. Fol-

lowing the crackdown in 2019, the number of cartels operating in municipalities with a gas pipeline increased, on average, by 17% (corresponding to an increase of 0.36 cartels). Similarly, Panel (a) of Figure 6 confirms that the average number of cartels in gas pipeline municipalities exhibited statistically significant growth after the crackdown. In 2019, there were 0.31 additional cartels in these areas relative to pre-crackdown 2018 levels. By 2021, there were 0.44 additional cartels.¹⁹ As shown in Panel (b), municipalities with refined oil pipelines also experienced increased cartel activity for two years following the crackdown, followed by a reduction in average cartel presence in 2021. Strong upward pre-trends in cartel activity in municipalities with refined oil pipelines was, along with thefts and homicides, an endogenous driver of the 2019 crackdown in these places.

Table 1: Crackdown Effects on Cartel Presence

	Number of Cartels			Cartel Presence (1/0)		
	All	Huachicol Specialized	Non-Huachicol Specialized	All	Huachicol Specialized	Non-Huachicol Specialized
	(1)	(2)	(3)	(4)	(5)	(6)
Gas pipeline $\times$ Post 2019	0.360*** (0.090)	0.230*** (0.048)	0.130** (0.060)	0.0339** (0.015)	0.0420*** (0.015)	0.0355** (0.016)
Refined oil pipeline $\times$ Post 2019	0.957*** (0.154)	0.282*** (0.071)	0.675*** (0.118)	-0.00678 (0.018)	-0.00735 (0.018)	0.0491** (0.021)
Observations	17311	17311	17311	17311	17311	17311
Adj. R-squared	0.884	0.819	0.867	0.642	0.640	0.606
Mean dep. var.	2.024	1.164	0.860	0.408	0.384	0.252

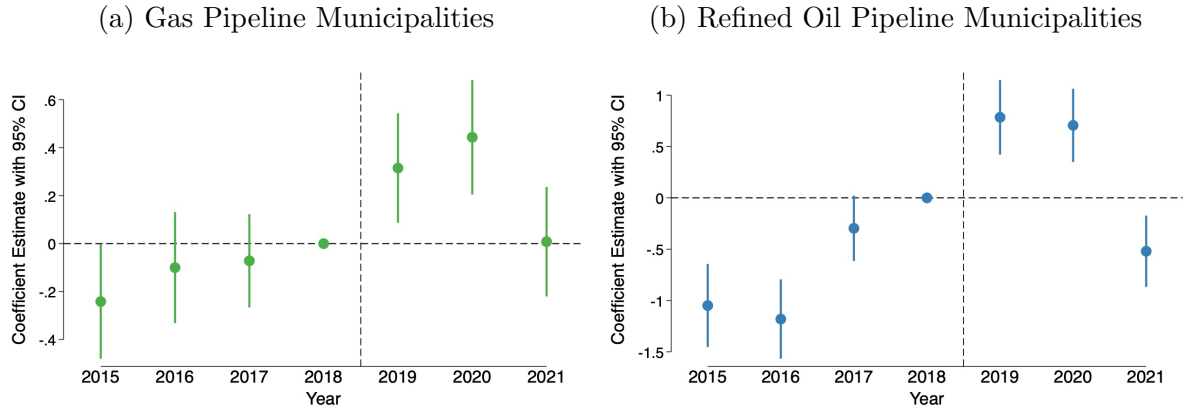
Note: This table reports coefficient estimates and standard errors from estimation of Equation 8. In Column (1) the dependent variable measures total active cartels; in column (2) the dependent variable measures fuel theft-specialized cartels, and in column (3) the dependent variable measures non-fuel theft specialized cartels, where fuel-theft specialized cartels are identified from [Etellekt Consultores \(2016\)](#), [Langner \(2017\)](#), [Castillo \(2021\)](#), and [González \(2020\)](#). Columns (4)-(6) report results for analogous binary categories to assess the extensive margin of cartel presence. Municipality and year-fixed effects are included in all specifications. Standard errors clustered at the municipality-level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

We next test the technological channel directly by comparing the crackdown response of cartels with and without prior specialization in fuel theft. Drawing on media coverage and a research report, we identify nine cartels that are most prominently involved in fuel theft ([Etellekt Consultores, 2016](#); [Langner, 2017](#); [Castillo, 2021](#); [González, 2020](#)).²⁰ Seven of these groups are large, well-established cartels responsible for 95% of all illegal fuel taps in 2016 ([Etellekt Consultores, 2016](#)), while the remaining two are smaller, local organizations also specialized in fuel theft ([Castillo, 2021](#); [González, 2020](#)). Column (2) of Table 1 reports estimated

¹⁹Considering the statistically insignificant but noteworthy upward pre-trend in cartel presence in gas pipeline municipalities in 6a, we assess robustness of the significant post-treatment effect using the method proposed by [Rambachan and Roth \(2023\)](#) and find that the observed pre-trend does not invalidate rejection of the null hypothesis for post-trend violations up to 1.2 times the magnitude of the pre-trend violation ($\bar{M} = 1.2$).

²⁰The nine cartels identified as *huachicoleros* are the Cartel Jalisco Nueva Generación (CJNG), Los Zetas, the Gulf Cartel, the Sinaloa Cartel, the Beltrán Leyva Organization, La Familia Michoacana, Los Caballeros Templarios, the Cartel de Santa Rosa de Lima (CSRL), and La Unión León. The last two began as smaller criminal groups and gained power through gasoline theft.

Figure 6: Dynamic Crackdown Effects on Cartel Presence



Note: Figure reports coefficient estimates and 95% confidence intervals for the number of active cartels in (a) municipalities hosting gas pipelines and (b) municipalities hosting refined oil pipelines, relative to non-pipeline municipalities, estimated using Equation 9. Specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

crackdown effects on the number of these fuel theft-specialized cartels operating in gas pipeline municipalities, while Column (3) reports corresponding estimates for non-specialized cartels. Consistent with Prediction 3, the crackdown was associated with an increase of 19.8% (0.23 additional cartels) in the number of fuel theft-specialized cartels in gas pipeline areas. This was a meaningfully larger response than that of non-specialized cartels, suggesting that cartels with previously accumulated experience in oil pipeline theft were more likely to substitute toward gas pipeline theft.²¹ Appendix Figure A5 reports analogous event study estimates, confirming that pre-crackdown trends in the presence of both types of cartels were parallel between gas pipeline and non-pipeline municipalities, and that the post-crackdown divergence is larger among specialized cartels.

6.3 Homicides

We now test whether cartel diversification into gas theft is associated with increased violence in affected municipalities. Table 2 shows results from estimation of Equation 8 on homicide rates for different demographic groups. Column (1) shows estimated changes in overall homicides. The crackdown was associated with an increase of 4.1 homicides per 100,000 inhabitants in municipalities with a gas pipeline, which is a 19% increase over the baseline mean of 21.3. In oil pipeline municipalities, we find a positive but statistically insignificant association with homicides. Due to non-parallel pre-trends for refined oil pipeline municipalities, we avoid drawing conclusions regarding the impact of the crackdown in these endogenously selected places.

²¹A Wald test comparing coefficient estimates in columns (2) and (3) yields a chi-squared value of 3.32 and p-value of 0.068, indicating a weakly significant statistical difference between effects for these two groups. Similar patterns hold in columns (5) and (6), which report effects on binary indicators of cartel presence. Individual results for each of the nine fuel theft cartels are reported in Appendix Table A4.

Columns (2)-(6) show pre/post crackdown changes across different demographic groups: males, females, Pemex employees, potentially cartel-related individuals, and military personnel. Estimates indicate that the post-crackdown rise in homicide rates in gas pipeline municipalities was driven by homicides of males (+18.5%) and potentially cartel-related individuals (+24.8%). In line with Prediction 5, these increases are likely the result of cartels fighting over territorial control of gas pipelines. We find no statistically significant associations of the crackdown with homicide rates among females, Pemex employees, or military members.

Table 2: Crackdown Effects on Homicides in Municipalities with Gas and Refined Oil Pipelines

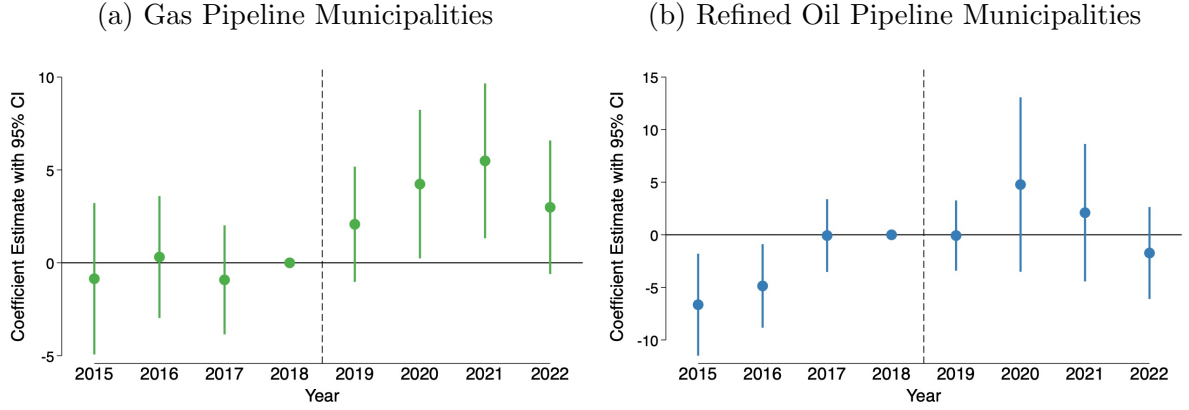
	All (1)	Male (2)	Female (3)	Pemex (4)	Cartel (5)	Military (6)
Gas pipeline $\times$ Post 2019	4.054*** (1.540)	3.491** (1.423)	0.355 (0.231)	-0.007 (0.011)	2.125** (0.886)	0.036 (0.059)
Refined oil pipeline $\times$ Post 2019	4.222 (2.658)	3.796 (2.437)	0.253 (0.308)	-0.018 (0.024)	1.408 (1.575)	0.054 (0.107)
Observations	19648	19648	19648	19648	19648	19648
Adj. R-squared	0.356	0.355	0.076	0.175	0.325	0.041
Mean dep. var.	21.31	18.87	2.30	0.02	8.58	0.05

Notes: This table reports coefficient estimates and standard errors from estimation of Equation 8. Dependent variables are measured as number of homicides per 100,000 inhabitants. Homicides of Pemex employees are estimated based on homicides of individuals of working age (18-60 years old) insured through Pemex. Potential cartel-related homicides are classified as males aged 18-40 killed by a firearm. Homicides of military personnel are estimated based on individuals insured through SEDENA, ages 18-60, and killed by a firearm. Municipality and year-fixed effects are included in all specifications. Standard errors clustered at the municipality-level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 7 reports event study results from estimation of Equation 9 for the overall homicide rate. Panel (a) of this figure confirms pre/post crackdown estimates. The crackdown was associated with a statistically significant increase of 2.1 homicides per 100,000 in the year following the intervention, and an increase of 4.8 homicides per 100,000 in 2021. We find no significant trends in homicide rates among gas pipeline municipalities prior to 2019, supporting the identifying parallel pre-trends assumption. Event study estimates of homicide rates disaggregated by demographic group are reported in Appendix Figure A7, confirming that the post-crackdown increase in homicide rates in gas pipeline municipalities was driven by homicides of men. Since men are more likely to be involved in organized crime and violence, this provides further evidence that conflict over control of gas pipelines was the underlying driver of the increase in homicides. Panel (b) illustrates dynamic associations between the crackdown and homicides in municipalities with refined oil pipelines. As with oil thefts, there is an upward trend in homicides prior to 2019, which is partly why the crackdown focused on these areas. Though

estimates are statistically insignificant, they suggest the crackdown may have halted the upward trend in homicide rates but did not substantially reduce violence in these places.

Figure 7: Dynamic Crackdown Effects on Homicide Rates

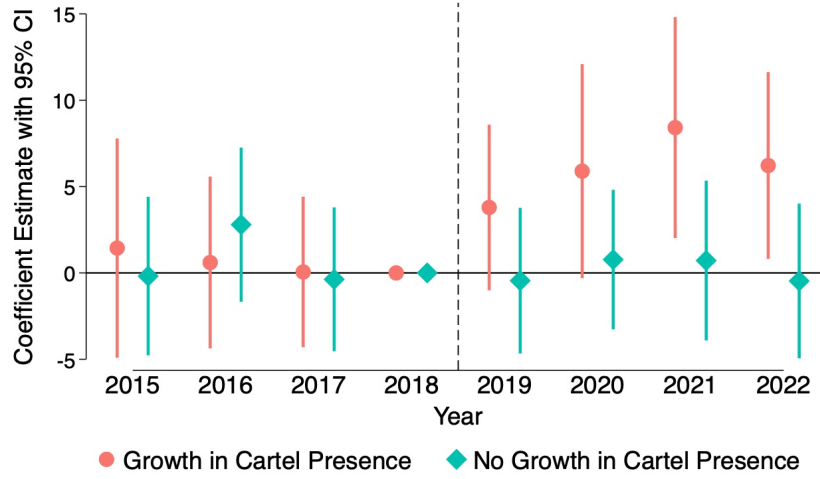


Note: Figure reports coefficient estimates and 95% confidence intervals for homicide rates per 100,000 residents in (a) municipalities hosting gas pipelines and (b) municipalities hosting refined oil pipelines, relative to non-pipeline municipalities, estimated using Equation 9. Specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

Prediction 5 further specifies that the homicide increase should be concentrated in municipalities that experienced cartel growth and are spatially proximate to oil pipeline infrastructure. Three heterogeneity analyses provide evidence consistent with this prediction. First, Appendix Figure A8 shows that the post-crackdown homicide increase was concentrated in gas pipeline municipalities with no recorded cartel presence during the 2015–2018 baseline period, consistent with conflict generated by the arrival of displaced cartels into previously uncontested territory. Second, Figure 8 splits gas pipeline municipalities into those where the number of active cartels grew after 2018 and those where it did not, finding that homicide growth was concentrated almost entirely in municipalities that experienced new cartel entry. Given that cartel growth is itself an endogenous post-treatment outcome, splitting the sample along this dimension does not provide a causal test of how cartel entry affects violence. It does, however, provide suggestive evidence for the mechanism. Third, Figure A9 shows that the homicide increase was concentrated in gas pipeline municipalities that also neighbor or host oil pipeline infrastructure, consistent with spatial substitution patterns in Prediction 2.

Finally, Appendix Figure A10 disaggregates homicide effects across gas pipeline municipalities that (i) also host an oil pipeline, (ii) neighbor a municipality with an oil pipeline, and (iii) neither neighbor nor host an oil pipeline. The results differ informatively from the analogous decomposition of gas thefts. Rather than concentrating in municipalities hosting both pipeline types, post-crackdown homicide rates increase most in municipalities neighboring oil pipeline areas, peaking with a slight delay in 2020. This pattern suggests a two-stage dynamic: cartels

Figure 8: Crackdown Effects on Homicides in Gas Pipeline Municipalities, by Cartel Growth



Note: Figure reports coefficient estimates and 95% confidence intervals for estimation of a modified version of Equation 9 with two treatment indicators, defined as (i) municipalities with gas pipelines where the number of active cartels grew by at least one after the beginning of the crackdown, and (ii) municipalities with gas pipelines that did not experience post-crackdown cartel growth. Specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

first diversified into gas theft within oil pipeline municipalities, where incumbency over existing territories suppressed conflict, before subsequently expanding gas tapping operations into neighboring municipalities. This spatial expansion, whether driven by crowding effects or interest in distancing operations from active enforcement, may have brought cartels into contact with rival groups or required coercion and intimidation of officials or opponents, leading to violence.²²

6.4 Other Crimes

Table A5 shows pre/post crackdown changes in other crimes: robberies, kidnappings, threats and extortion, and non-lethal violent crimes. In gas pipeline municipalities, robberies and threats and extortion do not change after the crackdown, but non-lethal violence increases by an average of 6.4%, or 6.1 incidents per 100,000 residents. Kidnappings decline by 0.28 cases per 100,000 residents, a reduction of 36.5% relative to the 2018 mean.²³ For municipalities hosting refined oil pipelines, robberies and non-lethal violence decreased by 24.1% (49.28 per 100,000

²²In Appendix Figure A11, we complement our homicide analysis with geocoded conflict event data from the Armed Conflict Location & Event Data Project (ACLED) as an alternative measure of violence. Huachicol-specialized clashes in oil pipeline municipalities increase (though not statistically significantly) at the onset of the crackdown (2018Q4), consistent with initial confrontations as fuel-theft organizations were displaced from oil pipeline corridors. In the following quarter (2019Q1), cartel clashes rise significantly in gas pipeline municipalities, consistent with displaced cartels encroaching into gas pipeline territories and generating violent competition there. These findings, drawn from an independent data source, corroborate our main homicide results and illustrate the sequentiality of cartel violence.

²³Self-reported data on other crimes should be interpreted with caution because people may be less inclined to report crimes in places where organized criminal groups are active. Thus, reporting rates could differ between pre and post-crackdown periods, as the rate of cartel presence increased. This reporting issue is not a concern for homicide data, which are collected from death certificates.

residents) and 14.7% (14.09 per 100,000 residents), respectively, which may be attributable to increased military and police presence from the crackdown.

In Table A6, we further decompose robberies by type: those targeting homes and businesses, highways and streets, machinery, and other. The crackdown was associated with reduced highway and street thefts in oil pipeline municipalities (-27.7%) but had no significant association with home and business robberies, consistent with increased military surveillance on roads rather than a broader reduction in criminal activity. Finally, thefts of machinery increased by 83% in gas pipeline municipalities, potentially reflecting the acquisition of equipment used for gas pipeline tapping operations.

6.5 Spatial Spillovers to Neighbors

Besides diversifying into gas pipeline theft, cartels affected by the 2019 crackdown may have shifted spatially into neighboring municipalities to avoid military operations around oil pipelines. To test this possibility, we redefine treated municipalities to be those that (i) neighbor a municipality hosting a refined oil pipeline or (ii) neighbor a municipality hosting a gas pipeline. We exclude all municipalities that host pipelines, leaving non-neighbor, non-pipeline municipalities as controls. DiD results on homicides based on this specification are reported in Table 3.

Estimates show no association of the crackdown with homicide rates in municipalities neighboring municipalities with gas pipelines. This is to be expected, since gas pipeline municipalities did not experience the military crackdown directly, and criminal groups operating in these areas were therefore not displaced. In contrast, we find large spillovers of violence into municipalities neighboring places with refined oil pipelines, with homicides increasing by 5.8 per 100 thousand residents, or 30%, in these places. Increased homicides are driven by homicides of men and potentially cartel-related individuals. This spillover effect suggests our main estimates for refined oil municipalities may underestimate the true impact of the crackdown on homicide rates, making our estimates conservative. Given that some municipalities neighbor both gas and refined oil pipeline municipalities, the presence of spatial spillovers could still bias our overall estimates by contaminating the non-pipeline control group. To verify this, we estimate a robustness check wherein we exclude direct pipeline neighbors from the control group (see Section 7).

7 Robustness Checks

In this section, we implement a variety of alternative specifications and tests to assess the robustness of our main results. Given spillovers to neighboring municipalities documented in Section 6.5, we first estimate a “spillover free” specification to ensure that spillovers of the

Table 3: Crackdown Effects on Homicides in Neighboring, Non-Pipeline Municipalities

	All (1)	Male (2)	Female (3)	Pemex (4)	Cartel (5)	Military (6)
Neighbor with gas pipeline $\times$ Post 2019	-1.303 (1.773)	-1.593 (1.647)	0.267 (0.323)	-0.00678 (0.007)	-0.621 (0.886)	0.00803 (0.027)
Neighbor with refined oil pipeline $\times$ Post 2019	5.806** (2.369)	5.631*** (2.125)	-0.0226 (0.449)	0.00294 (0.007)	2.091* (1.190)	0.0819 (0.068)
Observations	14352	14352	14352	14352	14352	14352
Adj. R-squared	0.327	0.331	0.0567	0.100	0.294	0.00214
Mean dep. var.	19.56	17.33	2.141	0.00461	7.570	0.0372

Notes: Table reports coefficient estimates and standard errors measuring spillovers of the 2019 crackdown onto homicide rates in municipalities neighboring municipalities that host refined oil or gas pipelines. The sample is restricted to municipalities with no refined oil or gas pipelines. The specification thus compares outcomes between non-pipeline neighbors (treated) and non-pipeline non-neighbors (controls). Dependent variables are measured as homicides per 100,000 inhabitants. Homicides of Pemex employees are estimated based on homicides of individuals of working age (18-60 years old) insured through Pemex. Potential cartel-related homicides are males aged 18-40, killed by a firearm. Homicides of military personnel are estimated based on individuals insured through SEDENA, ages 18-60, and killed by a firearm. Municipality and year-fixed effects are included in all specifications. Standard errors clustered at the municipality-level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

crackdown into neighboring municipalities do not bias our main estimates. To do so, we exclude all non-pipeline control municipalities that border places with refined oil or gas pipelines, leaving only non-pipeline, non-neighbor places as controls. Results, presented in Tables A7 and A8, show that our main estimates for homicides and cartel presence are strongly robust to this restriction. Point estimates of crackdown effects on homicides and cartel presence are slightly larger when excluding neighbors, suggesting that spillovers bias our main estimates downwards. The estimated effect of the crackdown on homicide rates in gas pipeline municipalities increases from +19% to +20%, and the estimated effect on cartel presence increases from +18% to +20% in the spillover-free specification.

Second, we address concerns that municipalities with pipeline infrastructure may have been chosen to host this infrastructure for endogenous reasons and may exhibit systematically different trends. To do so, we implement a Coarsened Exact Matching (CEM) procedure (Iacus et al., 2012) to restrict the sample of treated municipalities and control municipalities to those that match on pre-treatment characteristics, thus ensuring that treated and control municipalities are comparable. We match municipalities exactly on state and quintiles of pre-treatment (2015-2018) homicide rates, number of active cartels, formal employment share, population, and income level. We keep only municipalities that match and include matching weights in the regressions. Results, reported in Appendix Figure A12 and Table A9, are strongly robust to this restriction. The estimated impact on homicide rates in gas pipeline municipalities decreases from +19% to +18%, and the estimated effect on cartel presence decreases from +18% to +14% when restricting treated and control units to the matched subsample.

Third, we re-estimate DiD and event study specifications using log-transformed outcomes,

which are less influenced by extreme values. Log-transforming outcomes introduces difficulties in the presence of heteroskedasticity and large proportions of zero-value outcomes, as is our case (Manning and Mullahy, 2001; Silva and Tenreyro, 2006). In DiD models, this can result in level and log-transformed specifications yielding coefficient estimates with different signs (McConnell, 2023). Log-transformed results for homicides and cartel presence, reported in Appendix Tables A10 and A11, are very similar to our preferred estimates.

Fourth, we assess the extent to which our results are driven by CJNG territorial reorganization around fentanyl trafficking routes. We re-estimate cartel presence event studies excluding CJNG from the count and find that the post-crackdown increase in gas pipeline municipalities remains significant (Appendix Figure A13). Likewise, the cartel-level disaggregation in Appendix Table A4 shows significant growth in post-crackdown presence in gas pipeline municipalities for Sinaloa, Golfo, Juárez, and CSRL cartels in addition to CJNG, inconsistent with a story in which CJNG expansion alone drives our results. As an additional check, we re-estimate event studies on cartel presence and homicides controlling for baseline (2015–2018) CJNG presence interacted with year fixed effects, which at least partially captures CJNG’s territorial footprint after the 2013 reorganization. Results are robust to this additional control (Appendix Figures A14 and A15). Finally, we re-estimate event studies for excluding any municipality where a fentanyl seizure occurred between 2015-2021, using data obtained through a freedom of information request to SEDENA. Results are robust to this exclusion (Appendix Figures A16-A17).

Next, we address concerns that potentially differential Covid-19 infection rates in pipeline and non-pipeline municipalities from 2020 onward could confound our estimate of crackdown impacts on homicides and cartel presence. To do so, we collect municipality-year panel data on Covid-19 infection rates per 100,000 residents from Mexico’s Directorate General of Epidemiology and re-estimate our main specifications controlling for these rates. Results are strongly robust to this additional control (see Appendix Table A12). Event studies controlling for Covid-19 rates are also strongly robust (results available upon request).

To assess whether results could be driven by a single outlier state, we perform a leave-one-out analysis by re-estimating event studies for our main outcomes of interest (pipeline thefts, homicide rates, and cartel presence) repeatedly with one state omitted each time. We do so for each state that was treated by the crackdown, as identified in contemporary news coverage. Estimates, reported in Appendix Figure A18, are highly stable across leave-one-out specifications, suggesting results are not driven by a single outlier state.

Finally, we assess the robustness of event study estimates for pipeline thefts, homicides, and

cartel presence to potential violations of the identifying parallel pre-trends assumption using the methodology developed in [Rambachan and Roth \(2023\)](#). Robust confidence sets for different values of a parameter $\bar{M}$ – measuring the degree to which post-treatment trends may deviate from a linear extrapolation of pre-treatment trends – are reported in Appendix Figure [A19](#). Gas pipeline thefts exhibit an $\bar{M}$ breakdown value of 2, indicating that rejection of the null hypothesis for this outcome is robust to post-treatment violations of parallel trends up to 2 times the size of pre-treatment violations. The breakdown value of $\bar{M}$ for cartel presence is 1.2, indicating that the statistically insignificant but notable upward pre-trend for this outcome observed in Figure [6](#) does not invalidate rejection of the null hypothesis for post-trend violations up to 1.2 times the magnitude of the pre-trend violation. For homicides, the $\bar{M}$ breakdown value is 0.25, indicating that rejection of the null hypothesis is robust to violations of parallel trends that are slightly more than linear.

8 Discussion and Policy Implications

A growing literature documents how law enforcement crackdowns on organized crime generate unintended spillovers, as criminal organizations adapt by shifting activity toward less-policed alternatives. Prior work has examined these episodes in isolation, treating diversification as a one-shot lateral move in response to a single enforcement shock. In this paper, we take a broader view of criminal adaptation as sequential and self-reinforcing. Each enforcement shock prompts diversification into technologically and spatially adjacent sectors, and each diversification event accumulates durable organizational capital that lowers barriers to further adaptation. The result is a ratchet effect in which sector-specific crackdowns endogenously expand the criminal opportunity set, leaving cartels more capable, more diversified, and with a wider geographic footprint than before.

We document this dynamic empirically in the context of Mexico’s 2019 crackdown on oil pipeline theft. The crackdown was associated with a temporary reduction in oil thefts, but did not root out violence or cartel presence in targeted areas. Instead, it encouraged criminal organizations, especially those with prior experience in oil theft, to diversify into thefts from nearby gas pipelines. Theft of highly-pressurized gas products is a more technologically advanced operation than oil theft, requiring deeper infiltration of pipeline operators and frequently resulting in fires and explosions. Cartels’ shift into gas theft required them to expand geographically and compete over gas pipeline territories, plausibly contributing to a significant increase in cartel presence (+18%) in these areas. In turn, competition over pipelines was accompanied by a rise in violence, with homicide rates in gas pipeline municipalities growing by

19% after the crackdown – an effect we interpret as plausibly, though not definitively, causal, given potential confounders such as contemporaneous territorial conflicts. Cartels may have also responded to the crackdown on oil theft by expanding their reach into neighboring, non-pipeline municipalities, leading to a 30% increase in homicide rates in these places. When oil enforcement subsequently eased, oil theft recovered to 80% of its 2018 peak while gas theft remained elevated, consistent with the irreversibility prediction of our model.

These findings have implications for the design of policies aimed at combating organized crime and pipeline theft. Our model highlights two general insights. First, sector-specific crackdowns that leave adjacent sectors ungarded risk provoking adaptive diversification and spreading cartel presence more broadly. If crackdowns are undertaken, diversification into technologically and spatially related sectors should be anticipated and these sectors should be protected simultaneously. Second, policies that directly target cartel organizational capacity, rather than activity in a single sector, are likely to be more effective at reversing the ratchet. These policies could include institutional reforms to combat corruption or operations targeting cartel finances, which raise operating costs across all illicit sectors simultaneously, rather than merely displacing activity from one to another.

Specific to pipeline theft, precision policing, technology adoption, and institutional reforms are likely to be more effective and less disruptive than mass pipeline shutdowns. In Mexico, the military’s approach of shutting down pipelines and re-routing fuel shipments through tanker trucks led to spikes in local fuel prices, potentially incentivizing further thefts and eroding local support for enforcement efforts, while failing to address root issues such as poor monitoring of pipeline corridors, infiltration of pipeline operators, and lack of local economic opportunities (Crisis Group, 2022). Technologies to deter or rapidly detect pipeline thefts include ground pressure sensors, cameras and drones, and pipeline pressure monitors that can identify taps in real time (Atmos International, 2024; Adegboye et al., 2019). Deployment of these technologies could help catch thieves in the act or quickly shut off illegal taps, reducing the duration of profitable siphoning. Teams specialized in rapid response to tap detections could be drawn from federal personnel and rotated between locations, making infiltration more difficult. Policing efforts can be focused on stretches of pipeline closed for maintenance, when they are most vulnerable to tapping. Oil and gas theft may also be combated on the retail end by monitoring and auditing gas stations and LPG distributors, which often buy and resell stolen fuel. Ultimately, governments must maintain control over areas hosting energy infrastructure and promote formal economic opportunities for residents such that criminal groups struggle to take hold — potentially by channeling a share of oil and gas revenues toward communities hosting

pipeline infrastructure to create shared incentives for protection and maintenance.

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1 Theory Appendix

1.1 Transferability Coefficients

Table A1 summarizes the transferability coefficients implied by Assumption 1 for the three sectors in our empirical setting. High transferability from oil to gas theft reflects overlapping institutional features, technical methods, and geographic footprints, while low transferability from drug trafficking to either pipeline sector reflects the distinct institutional relationships, logistics, and territorial requirements of each activity.

Table A1: Transferability Coefficients Across Sectors, Based on Assumption 1

Transition	κ^τ (Technological)	κ^σ (Spatial)	Overall
$D \rightarrow O$	Low	Low	Low
$D \rightarrow G$	Very low	Low	Very low
$O \rightarrow G$	High	High	High

Assumption 1 is calibrated to match the empirical record, i.e., that cartels diversified sequentially from drug trafficking into oil theft and then into gas theft. However, the framework is general. Any distance structure can be imposed to analyze diversification into other illicit sectors, with the predicted diversification path emerging endogenously from assumed coefficients. We illustrate this with two additional cases drawn from the Mexican context.

From drug trafficking to avocado extortion. De Haro (2025) documents cartel diversification into extortion of avocado farmers in Michoacán following a negative shock to drug trafficking revenues. In our framework, this transition is characterized by high spatial transferability – drug trafficking and avocado farming both concentrated in Michoacán, so cartels’ existing territorial presence transferred directly – and high technological transferability, as agricultural extortion requires little technical expertise beyond the political relationships and corrupted enforcement officials already accumulated through drug operations. The model therefore predicts rapid diversification into avocado extortion following a drug revenue shock, consistent with the empirical record.

From traditional drug trafficking to fentanyl. López Cruz and Torrens (2026) document that the US fentanyl demand shock starting around 2013 pushed Pacific-based Mexican cartels to re-optimize their trafficking routes, generating a second wave of violence in municipalities largely unaffected by the first wave of the War on Drugs. In our framework, this pattern reflects moderate transferability from traditional drug trafficking to fentanyl. Technologically, fentanyl shares institutional features with traditional drug trafficking (e.g., distribution networks, customer relationships, corruption of federal officials) but requires different chemical precursor supply chains and production equipment. Spatially, fentanyl trafficking relies on Pacific ports and east-west highway corridors rather than traditional north-south drug routes, so transferability was highest for cartels already operating along the Pacific coast –whose existing territorial presence overlapped most closely with the ports through which fentanyl precursors flow. The model therefore predicts that diversification into fentanyl should concentrate among

Pacific-based cartels and generate violence in newly contested municipalities, consistent with [López Cruz and Torrens's](#) findings.

Together, the cases of cartel diversification into avocado extortion and fentanyl trafficking illustrate that our model accommodates diverse diversification patterns and generalizes beyond enforcement crackdowns to any sectoral revenue shock.

1.2 Derivation of Entry Thresholds

Stage 1: Adding sector j to a single-sector portfolio

With only D active and the capacity constraint binding at M_0 , $s_D + s_j = M_0$, so $s_D = M_0 - s_j$. Total profit as a function of s_j is:

$$\Pi(s_j) = \pi_D(M_0 - s_j) - \frac{\gamma}{2}(M_0 - s_j)^2 + \pi_j s_j - \frac{\gamma}{2}s_j^2. \quad (10)$$

Differentiating $\Pi(s_j)$ with respect to s_j :

$$\frac{d\Pi}{ds_j} = -\pi_D + \gamma(M_0 - s_j) + \pi_j - \gamma s_j \quad (11)$$

$$= \pi_j - (\pi_D - \gamma M_0) - 2\gamma s_j. \quad (12)$$

Before sector j is added, only D is active and the constraint binds at $s_D = M_0$. The first-order condition for D then pins down the shadow price: $s_D^* = (\pi_D - \lambda_1)/\gamma = M_0$, so $\lambda_1 = \pi_D - \gamma M_0$. Substituting into the expression above:

$$\frac{d\Pi}{ds_j} = (\pi_j - \lambda_1) - 2\gamma s_j. \quad (13)$$

The factor 2γ reflects two effects. First, coordination costs in j rise at rate γ as s_j increases. Second, since the constraint binds, each additional unit of s_j requires removing a unit from s_D , and as s_D shrinks its marginal value rises, contributing a further γ . The marginal return therefore starts at $\pi_j - \lambda_1$ and declines to zero at:

$$(\pi_j - \lambda_1) - 2\gamma s_j = 0 \quad (14)$$

$$s_j^* = \frac{\pi_j - \lambda_1}{2\gamma}. \quad (15)$$

The profit gain from adding sector j is the area under the marginal return curve from $s_j = 0$ to $s_j = s_j^*$, that is, the total gain from incrementally reallocating capacity from D to j . Since the marginal return $(\pi_j - \lambda_1) - 2\gamma s_j$ is linear in s_j , starting at $\pi_j - \lambda_1$ and declining to zero at s_j^* , this area is a triangle with base s_j^* and height $\pi_j - \lambda_1$:

$$\Delta\Pi_j^1 = \frac{1}{2} \cdot s_j^* \cdot (\pi_j - \lambda_1) = \frac{1}{2} \cdot \frac{\pi_j - \lambda_1}{2\gamma} \cdot (\pi_j - \lambda_1) = \frac{(\pi_j - \lambda_1)^2}{4\gamma}. \quad (16)$$

The cartel enters j if and only if $\Delta\Pi_j^1 \geq \tilde{F}_j(K_D)$:

$$\frac{(\pi_j - \lambda_1)^2}{4\gamma} \geq \tilde{F}_j(K_D) \implies (\pi_j - \lambda_1) \geq 2\sqrt{\gamma \tilde{F}_j(K_D)} \implies \pi_j \geq \lambda_1 + 2\sqrt{\gamma \tilde{F}_j(K_D)} \equiv \bar{\pi}_j. \quad (17)$$

Threshold $\bar{\pi}_j$ is the minimum per-unit net revenue that makes entry into j worthwhile given shadow price λ_1 and effective entry cost $\tilde{F}_j(K_D)$.

Stage 2: Adding sector j to a two-sector portfolio

In the context of Stage 2, $j = G$ and the two incumbent sectors are D and O . With both active and the constraint binding at M_1 , the optimal scales satisfy $s_D^* + s_O^* = M_1$, with shadow price:

$$\lambda_2 = \frac{\pi_D + \pi_O}{2} - \frac{\gamma M_1}{2}. \quad (18)$$

Now consider adding j at scale s_j . The remaining capacity $M_1 - s_j$ is optimally reallocated between D and O . By the envelope theorem, the opportunity cost of diverting ds_j units of capacity away from the $\{D, O\}$ subproblem equals the shadow price of that subproblem, μ , which depends on s_j :

$$\mu(s_j) = \frac{\pi_D + \pi_O}{2} - \frac{\gamma(M_1 - s_j)}{2} = \lambda_2 + \frac{\gamma s_j}{2}. \quad (19)$$

The marginal return to increasing s_j is therefore:

$$\frac{d\Pi}{ds_j} = \pi_j - \gamma s_j - \mu(s_j) \quad (20)$$

$$= \pi_j - \gamma s_j - \lambda_2 - \frac{\gamma s_j}{2} \quad (21)$$

$$= (\pi_j - \lambda_2) - \frac{3\gamma}{2} s_j. \quad (22)$$

Compared to Stage 1, the marginal return declines at rate $\frac{3\gamma}{2}$ rather than 2γ , since the two incumbent sectors share the burden of releasing capacity to j : each cedes only $\frac{1}{2}$ unit per unit of s_j added, so the rise in opportunity cost is $\frac{\gamma}{2}$ rather than γ . Setting the marginal return to zero:

$$(\pi_j - \lambda_2) - \frac{3\gamma}{2} s_j = 0 \quad (23)$$

$$s_j^* = \frac{2(\pi_j - \lambda_2)}{3\gamma}. \quad (24)$$

The profit gain is again the area of the triangle:

$$\Delta\Pi_j^2 = \frac{1}{2} \cdot s_j^* \cdot (\pi_j - \lambda_2) = \frac{1}{2} \cdot \frac{2(\pi_j - \lambda_2)}{3\gamma} \cdot (\pi_j - \lambda_2) = \frac{(\pi_j - \lambda_2)^2}{3\gamma}. \quad (25)$$

For $j = G$, this gives $\Delta\Pi_G^2 = \frac{(\pi_G - \lambda_2)^2}{3\gamma}$.

General expression for n incumbent sectors

With n incumbent sectors active under binding constraint M , the shadow price of the incumbent subproblem when capacity $M - s_j$ is available to them is:

$$\mu(s_j) = \frac{\sum_{i=1}^n \pi_i}{n} - \frac{\gamma(M - s_j)}{n} = \lambda_n + \frac{\gamma s_j}{n}, \quad (26)$$

where $\lambda_n = \frac{\sum_{i=1}^n \pi_i}{n} - \frac{\gamma M}{n}$ is the shadow price with n active sectors. The marginal return to adding sector j is:

$$\frac{d\Pi}{ds_j} = \pi_j - \gamma s_j - \mu(s_j) \quad (27)$$

$$= (\pi_j - \lambda_n) - \gamma \left(1 + \frac{1}{n}\right) s_j \quad (28)$$

$$= (\pi_j - \lambda_n) - \frac{\gamma(n+1)}{n} s_j. \quad (29)$$

Setting to zero:

$$s_j^* = \frac{n(\pi_j - \lambda_n)}{\gamma(n+1)}. \quad (30)$$

The profit gain is:

$$\Delta\Pi_j^n = \frac{1}{2} \cdot s_j^* \cdot (\pi_j - \lambda_n) = \frac{n(\pi_j - \lambda_n)^2}{2\gamma(n+1)}. \quad (31)$$

For $n = 1$ this gives $\frac{(\pi_j - \lambda_1)^2}{4\gamma}$, confirming the Stage 1 expression. For $n = 2$ it gives $\frac{(\pi_j - \lambda_2)^2}{3\gamma}$, confirming the Stage 2 expression.

1.3 Proof of Proposition 1

The gap between thresholds is:

$$\bar{\pi}_G - \bar{\pi}_O = \left(\lambda_1 + 2\sqrt{\gamma \tilde{F}_G(K_D)} \right) - \left(\lambda_1 + 2\sqrt{\gamma \tilde{F}_O(K_D)} \right) = 2\sqrt{\gamma} \left(\sqrt{\tilde{F}_G(K_D)} - \sqrt{\tilde{F}_O(K_D)} \right) > 0, \quad (32)$$

which is positive because $\tilde{F}_G(K_D) > \tilde{F}_O(K_D)$. With the proportional entry cost specification, $\tilde{F}_G(K_D) = F_G[1 - \psi(\kappa_{DG}^\tau, \kappa_{DG}^\sigma)]$ and $\tilde{F}_O(K_D) = F_O[1 - \psi(\kappa_{DO}^\tau, \kappa_{DO}^\sigma)]$. Since $F_G > F_O$ and $\kappa_{DG}^\tau < \kappa_{DO}^\tau$ under Assumption 1 (so that $\psi(\kappa_{DG}^\tau, \kappa_{DG}^\sigma) < \psi(\kappa_{DO}^\tau, \kappa_{DO}^\sigma)$, meaning gas receives a smaller capital subsidy from drug experience than oil does), both effects push $\tilde{F}_G(K_D)$ above $\tilde{F}_O(K_D)$. The interval $(\bar{\pi}_O, \bar{\pi}_G]$ is therefore non-empty. Since both thresholds are decreasing in e_D^1 through λ_1 , a more severe crackdown shifts both downward by the same amount, preserving the gap. $\square$

1.4 Sufficient Condition for Proposition 4

Proposition 4 claims that there exist parameter values under which $\bar{\pi}_G^2 < \bar{\pi}_G^1$, so that gas entry becomes profitable in Stage 2 even though it was blocked in Stage 1. Since $\bar{\pi}_G^1 = \lambda_1 +$

$2\sqrt{\gamma \tilde{F}_G(K_D)}$ and $\bar{\pi}_G^2 = \lambda_2 + \sqrt{3\gamma \tilde{F}_G(K_O)}$, the condition $\bar{\pi}_G^2 < \bar{\pi}_G^1$ is equivalent to:

$$(\lambda_1 - \lambda_2) + \left(2\sqrt{\gamma \tilde{F}_G(K_D)} - \sqrt{3\gamma \tilde{F}_G(K_O)} \right) > 0. \quad (33)$$

The first term captures the *capacity channel*: $\lambda_1 - \lambda_2 > 0$ because both the oil crackdown (reducing π_O) and the expanded capacity $M_1 > M_0$ put downward pressure on λ_2 relative to λ_1 . The second term captures the *capital channel*: oil experience reduces the effective entry cost into gas, so $\tilde{F}_G(K_O) < \tilde{F}_G(K_D)$, which pushes the second term positive. Each channel individually shifts $\bar{\pi}_G^2$ below $\bar{\pi}_G^1$; condition (33) holds whenever their combined effect is sufficiently large.

1.5 Implications for Combating Organized Crime

Two potential policy levers for combating organized crime emerge from the model's structure.

Attacking the ratchet mechanism directly. Sector-specific enforcement raises $\phi_i(e_i)$, reducing scale s_i^* without affecting the channels through which the ratchet operates: sector-specific capital K_i and organizational capacity M . To sever the ratchet, enforcement must target both. Degrading K_i requires targeting the institutional infrastructure of criminal operations rather than their outputs: prosecuting and replacing corrupted energy officials degrades K_i^r ; dismantling territorial networks degrades K_i^σ . If K_i is destroyed, the capital subsidy in equation (2) disappears and effective entry costs rise back toward greenfield levels. Constraining M could involve financial operations targeting money laundering, asset seizures, or prosecution of senior management. However, the latter must be approached with caution: Mexico's experience with the kingpin strategy during the War on Drugs illustrates how targeting cartel leadership can fragment organizations into smaller groups, potentially reducing any single cartel's M while leaving aggregate criminal capacity intact or even increased, along with large increases in violence. These interventions nonetheless represent the most direct route to undoing the irreversibility in Corollary 1.

Simultaneous multi-sector enforcement. Finite enforcement resources, captured by $\sum_i e_i \leq E$, mean that concentrating effort on one sector necessarily leaves others less policed, creating opportunities for diversification. Spreading enforcement across sectors avoids this, but reduces intensity in any particular sector. When enforcement is strong enough that the capacity constraint does not bind ($\lambda = 0$), total scale $\sum_i s_i^* = \sum_i \pi_i(e_i)/\gamma$ is directly decreasing in each $\phi_i(e_i)$, so minimizing $\sum_i s_i^*$ is equivalent to maximizing $\sum_i \phi_i(e_i)$ subject to the budget constraint.

Proposition 6 (Optimal Multi-Sector Enforcement). *When the capacity constraint does not bind ($\lambda = 0$), the government minimizes total criminal activity by equalizing the marginal deterrence effect of enforcement across all active sectors:*

$$\phi'_i(e_i) = \phi'_j(e_j) \quad \text{for all active } i, j. \quad (34)$$

Concentrating enforcement on a single sector is suboptimal whenever $\phi''_i < 0$: diminishing

*marginal deterrence means reallocating some effort toward less-policed sectors reduces total criminal activity more than concentrating it on one.*²⁴

Proof. The government maximizes $\sum_i \phi_i(e_i)$ subject to $\sum_i e_i \leq \bar{E}$. The Lagrangian yields first-order condition $\phi'_i(e_i) = \mu$ for all active i , where μ is the shadow price of the enforcement budget. Since $\phi''_i \leq 0$, ϕ'_i is non-increasing, so any solution with unequal enforcement levels could be improved by reallocating from the over-enforced sector (where ϕ'_i is low) to the under-enforced one (where ϕ'_i is high), until the equalization condition holds. $\square$

1.6 The Ratchet Effect Under Capital Depreciation

In this section, we discuss how relaxing assumptions about the durability of sector specific capital and organizational capacity would affect Propositions 2 through 5 and Corollary 1.

In the baseline model, K_i is a binary indicator that equals one if the cartel has accumulated sector-specific capital in sector i . Consider instead a continuous, depreciating capital stock for each sector i , evolving according to $K_i^t = (1 - \delta)K_i^{t-1} + g(s_i^t)$, where $\delta \in (0, 1]$ is the depreciation rate and the accumulation function $g(\cdot)$ satisfies $g'(s_i^t) > 0$, $g''(s_i^t) \leq 0$, and $g(0) = 0$. Under this specification, capital decays geometrically toward zero when activity ceases ($s_i^t = 0$) and can be maintained or increased when activity is high enough that $g(s_i^t) \geq \delta K_i^{t-1}$. Further, suppose that M depends on the continuous K_i^t . Because M caps total activity across sectors as a function of available capital, a decline in one sector's capital tightens the feasible set for activity in all other sectors.

Following these relaxations of our baseline assumptions, Proposition 2 remains unaffected, as it is a derived result following from the interior-solution logic of the cartel's optimization problem within a single stage. In Stage 2, the crackdown on oil theft lowers the potential revenue from that sector, leading the cartel to reduce its scale of operations, s_O . Since capital now depends on the scale at which the cartel previously operated, this decline causes K_O – and therefore M_2 – to decay, reducing the capital subsidy to gas entry (by raising $\tilde{F}$) and raising the opportunity cost of adding gas (by shrinking organizational capacity M and raising the shadow price λ). Therefore, at a sufficient rate of decay, both the capacity and capital channels can vanish, and the gas entry threshold converges to what it would have been without any oil experience, eliminating the ratchet effect (Proposition 3). Without a ratchet effect to lower $\bar{\pi}_G^2$ below $\bar{\pi}_G^1$ (Proposition 4), gas entry would not become newly profitable in Stage 2, and sequential crackdowns would no longer endogenously expand the cartel's opportunity set.

Proposition 5 (Persistence of Diversification) states that when oil enforcement eases, the cartel scales up oil operations while gas theft remains active as long as gas is profitable. Under the depreciating-capital specification, even if oil capital K_O (and hence organizational capacity M) partially decayed while s_O was suppressed, both K_O and M rebuild once s_O recovers. Thus, temporary enforcement does not permanently undo the persistence of diversification. Only enforcement that keeps s_O suppressed indefinitely would prevent capital and capacity from recovering.

²⁴When the capacity constraint binds ($\lambda > 0$), total scale equals M regardless of how enforcement is allocated across sectors. Reducing total criminal activity then requires concentrating enforcement sufficiently on one sector to force exit entirely, pushing $\pi_i(e_i) \leq \lambda$, so that freed capacity cannot be redeployed if no adjacent sector is available.

Corollary 1 (Irreversibility of Diversification) is more sensitive to sustained depreciation. In the model, no enforcement policy that operates solely through $\phi_i(e_i)$ can undo diversification: sector-specific capital and organizational capacity are permanent, so once the cartel has entered gas, this sector remains in its portfolio. With depreciating capital, this irreversibility becomes conditional. If enforcement is strong and persistent enough to permanently raise $\phi_i(e_i)$ in a sector – keeping s_i low – capital in that sector depreciates, shrinking M . As M shrinks, the cartel reallocates toward its highest-return sectors and may withdraw entirely from higher-cost sectors, gradually de-diversifying.

Summing up, Propositions 3 and 4 attenuate smoothly as capital depreciates, while Proposition 5 is robust to temporary depreciation and Corollary 1 depends on whether depreciation is sustained long enough to permanently shrink M . The purpose of our model is to demonstrate how the ratchet mechanism *can* generate persistent, expanding diversification, not that it must. Mexico’s experience of cartel diversification from drug trafficking into oil theft and then gas theft illustrates how this dynamic can play out, offering a cautionary tale and suggesting model assumptions align with reality in this context. However, different assumptions about capital depreciation can change the outcome: enforcement that directly and persistently attacks organizational capacity, or that operates simultaneously across multiple sectors rather than one at a time, is effectively equivalent to the depreciation analyzed here, showing how diversification can be reversed rather than ratcheted up.

2 Supplementary Tables

2.1 Pipeline Thefts and World Fuel Prices

Using monthly data on fuel prices from CRE, we regress the number of oil or gas pipeline thefts recorded in a municipality-month on (i) national gasoline and diesel prices, (ii) national LPG prices, and (iii) Brent Crude world oil prices, including municipality fixed effects and clustering standard errors at the municipality level. Results, reported in Table A2, reveal a significant positive association between fuel prices and thefts. A 10% increase in exogenous world oil prices is associated with a 3.6% rise in refined oil pipeline thefts. Theft is even more responsive to (less plausibly exogenous) national prices: a 10% increase in Mexican gasoline and diesel prices is associated with a 5% increase in fuel thefts. If efforts to crack down on pipeline theft lead to increased fuel prices, this could paradoxically encourage additional thefts.

Table A2: Sensitivity of Gas and Refined Oil Thefts to Fuel Prices

		Dependent Variable	
		Illegal Oil Taps	Illegal Gas Taps
		(1)	(2)
<i>(a) Mexican Gasoline & Diesel Price</i>	<i>Coef.</i>	0.520***	0.107***
	<i>St. Err.</i>	(0.072)	(0.025)
<i>(b) Mexican LPG Price</i>	<i>Coef.</i>	0.132**	0.022
	<i>St. Err.</i>	(0.058)	(0.027)
<i>(c) Brent World Oil Price</i>	<i>Coef.</i>	0.378***	0.069***
	<i>St. Err.</i>	(0.052)	(0.019)

Note: Table presents coefficient estimates and standard errors in parentheses for regressions of the number of illegal taps on the country-level mean retail price of gasoline and diesel (Panel a), the country-level mean liquid petroleum gas (LPG) retail price (Panel b), and Brent Crude world oil price (Panel c), which moves exogenously to local events in Mexico. Column (1) reports results for illegal taps on refined oil pipelines, while Column (2) reports results for illegal taps on gas pipelines, both measured at the municipality-month level. All regressions include municipality fixed effects and standard errors are clustered at the municipality level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A3: Summary Statistics in Municipalities with Gas or Refined Oil Pipelines

	Gas or refined oil			Rest of country		
	Mean	SD	Obs.	Mean	SD	Obs.
Reports of illegal taps						
Gas	0.24	1.87	2,660	0.00	0.12	7,232
Refined petroleum	12.88	46.79	2,660	0.34	5.44	7,232
Homicide rates						
All	23.13	30.07	2,648	18.97	38.01	7,176
Male	20.58	27.79	2,648	16.87	34.60	7,176
Female	2.44	4.69	2,648	2.03	7.73	7,176
Cartel related	9.86	16.89	2,648	7.19	19.00	7,176
Pemex employee	0.06	0.62	2,648	0.01	0.16	7,176
Military	0.05	0.65	2,648	0.02	0.41	7,176
Cartel data						
All cartels	3.22	5.42	2,660	1.18	3.00	7,232
Huachicol specialized	1.83	2.38	2,660	0.71	1.50	7,232
Non-huachicol specialized	1.39	3.43	2,660	0.48	1.75	7,232
Robberies						
All	384.71	404.53	2,648	138.46	235.58	7,176
Home & business	93.35	122.41	2,648	31.75	68.02	7,176
Highway & street	160.71	221.97	2,648	50.07	109.61	7,176
Machinery	1.43	4.98	2,648	0.79	5.20	7,176
Other	114.38	135.82	2,648	47.22	96.01	7,176
Other crimes						
Kidnapings	1.49	4.15	2,648	0.64	3.64	7,176
Threats & extortion	50.98	69.29	2,648	25.75	58.75	7,176
Non-lethal violence	125.52	121.88	2,648	65.93	102.83	7,176
Socioeconomic Indicators						
Population (in thousands)	110.39	232.68	2,648	21.88	45.13	7,176
Formal employment	0.91	1.10	2,648	0.91	2.40	7,176
Higher Education Enrollment	0.03	0.04	670	0.03	0.04	829
Gini (2015)	0.37	0.03	656	0.38	0.04	1,790

Notes: This table presents the summary statistics of the relevant variables used in this study. Homicide, robbery, and crime rates are expressed as the number of cases per 100,000 people. Huachicol specialized cartels refer to the ten criminal organizations identified as having high participation in fuel theft. Non-lethal violence consists of injuries and other crimes that threaten life. Information on higher education enrollment from ANUIES was not available for 2015-2016. Data on gini was obtained from the 2015 census.

Table A4: Crackdown Effects on Cartel Presence in Municipalities, Disaggregated by Cartel

	CJNG (1)	Zetas (2)	Golfo (3)	Sinaloa (4)	BLO (5)
Gas pipeline $\times$ Post 2019	0.0609*** (0.016)	0.0164 (0.013)	0.0341*** (0.013)	0.0495*** (0.013)	0.00940 (0.009)
Refined Oil pipeline $\times$ Post 2019	-0.00888 (0.019)	0.000793 (0.018)	0.0546*** (0.020)	0.00126 (0.018)	0.0500*** (0.015)
Observations	17311	17311	17311	17311	17311
Adj. R-squared	0.624	0.550	0.540	0.566	0.596
Mean dep. var.	0.273	0.178	0.124	0.172	0.101

	Fam. Mich. (6)	Templarios (7)	Juarez (8)	CSRL (9)	Union Leon (10)
Gas pipeline $\times$ Post 2019	0.00957 (0.010)	0.00431 (0.009)	0.0148* (0.008)	0.0302** (0.012)	0.000972 (0.003)
Refined Oil pipeline $\times$ Post 2019	0.0206 (0.018)	0.0260 (0.017)	0.00420 (0.014)	0.116*** (0.020)	0.0167*** (0.006)
Observations	17311	17311	17311	17311	17311
Adj. R-squared	0.571	0.556	0.536	0.470	0.439
Mean dep. var.	0.108	0.0941	0.0532	0.0577	0.00335

Note: This table reports coefficient estimates and standard errors from estimation of Equation 8. The dependent variables are dichotomous variables equal to one if the respective cartel is present in a municipality in a year. CJNG stands for Cartel Jalisco Nueva Generación, BLO for Beltrán Leyva Organization, Fam. Mich. for La Familia Michoacana, and CSRL for Cartel Santa Rosa de Lima. Clustered standard errors at the municipality level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Crackdown Effects on Other Crimes

	Robberies (1)	Kidnap (2)	Threats & Extortion (3)	Non-lethal Violence (4)
Gas pipeline $\times$ Post 2019	0.722 (8.472)	-0.287* (0.151)	3.281 (2.869)	6.105** (3.021)
Refined oil pipeline $\times$ Post 2019	-49.28*** (12.926)	-0.345 (0.244)	0.228 (3.999)	-14.09*** (4.249)
Observations	19648	19648	19648	19648
Adj. R-squared	0.838	0.119	0.526	0.736
Mean dep. var.	204.4	0.786	49.83	95.57

Note: Table reports coefficient estimates and standard errors from estimation of Equation 8. Dependent variables are measured as number of crimes per 100,000 inhabitants. Non-lethal violence consists of injuries and other crimes that threaten life. Municipality and year-fixed effects are included in all specifications. Standard errors clustered at the municipality-level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A6: Crackdown Effects on Robberies (Disaggregated)

	Home & Business (1)	Highway & Street (2)	Machinery (3)	Other (4)
Gas pipeline $\times$ Post 2019	-1.429 (2.878)	1.026 (4.714)	1.046*** (0.283)	2.201 (3.408)
Refined oil pipeline $\times$ Post 2019	-6.933 (4.913)	-22.45*** (7.569)	-0.521 (0.366)	-18.22*** (5.017)
Observations	19648	19648	19648	19648
Adj. R-squared	0.752	0.792	0.146	0.740
Mean dep. var.	48.13	80.80	1.255	65.83

Note: Table reports coefficient estimates and standard errors from estimation of Equation 8. Dependent variables are measured as number of robberies of specific types per 100,000 inhabitants. Municipality and year-fixed effects are included in all specifications. Standard errors clustered at the municipality-level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A7: No-Neighbors Analysis: Crackdown Effects on Homicides

	All (1)	Male (2)	Female (3)	Pemex (4)	Cartel (5)	Military (6)
Gas pipeline $\times$ Post 2019	4.329*** (1.555)	3.756*** (1.439)	0.341 (0.234)	-0.007 (0.011)	2.192** (0.895)	0.045 (0.059)
Refined oil pipeline $\times$ Post 2019	4.359 (2.663)	3.928 (2.442)	0.246 (0.309)	-0.018 (0.025)	1.442 (1.578)	0.058 (0.107)
Observations	18088	18088	18088	18088	18088	18088
Adj. R-squared	0.365	0.360	0.077	0.177	0.321	0.050
Mean dep. var.	21.29	18.88	2.27	0.02	8.59	0.05

Note: This table reports coefficient estimates and standard errors from Equation 8. This specification drops all non-pipeline municipalities that are adjacent to a municipality with a refined oil or gas pipeline to avoid bias introduced by potential spillovers into these control units. Dependent variables are measured as number of homicides per 100,000 inhabitants. Homicides of Pemex employees are estimated based on homicides of individuals of working age (18-60 years old) insured through Pemex. Potential cartel-related homicides are classified as males aged 18-40 killed by a firearm. Homicides of military personnel are estimated based on individuals insured through SEDENA, ages 18-60, and killed by a firearm. Municipality and year-fixed effects are included in all specifications. Standard errors clustered at the municipality-level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A8: No-Neighbors Analysis: Crackdown Effects on Cartel Presence

	Number of Cartels			Cartel Presence (1/0)		
	All	Huachicol Specialized	Non-Huachicol Specialized	All	Huachicol Specialized	Non-Huachicol Specialized
	(1)	(2)	(3)	(4)	(5)	(6)
Gas pipeline $\times$ Post 2019	0.405*** (0.091)	0.258*** (0.048)	0.147** (0.060)	0.0412*** (0.015)	0.0497*** (0.015)	0.0425*** (0.016)
Refined oil pipeline $\times$ Post 2019	0.988*** (0.155)	0.300*** (0.071)	0.687*** (0.119)	-0.00287 (0.018)	-0.00327 (0.018)	0.0531** (0.021)
Observations	15827	15827	15827	15827	15827	15827
Adj. R-squared	0.888	0.825	0.872	0.646	0.644	0.618
Mean dep. var.	2.057	1.166	0.890	0.405	0.380	0.252

Note: This table reports coefficient estimates and standard errors from Equation 8. This specification drops all non-pipeline municipalities that are adjacent to a municipality with a refined oil or gas pipeline to avoid bias introduced by potential spillovers into these control units. In Column (1) the dependent variable measures total active cartels; in column (2) the dependent variable measures fuel theft-specialized cartels, and in column (3) the dependent variable measures non-fuel theft specialized cartels, where fuel-theft specialized cartels are identified from [Etelekt Consultores \(2016\)](#), [Langner \(2017\)](#), [Castillo \(2021\)](#), and [González \(2020\)](#). Columns (4)-(6) report results for analogous binary categories to assess the extensive margin of cartel presence. Municipality and year-fixed effects are included in all specifications. Standard errors clustered at the municipality-level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A9: Results on Matched-subsample

	Homicides			Number of Cartels		
	All	Male	Cartel	All	Huachicol Specialized	Non-Huachicol Specialized
	(1)	(2)	(3)	(4)	(5)	(6)
Gas pipeline $\times$ Post 2019	3.911** (1.600)	3.424** (1.479)	2.049** (0.919)	0.285*** (0.092)	0.207*** (0.052)	0.078 (0.055)
Refined oil pipeline $\times$ Post 2019	3.936 (2.925)	3.612 (2.689)	1.249 (1.742)	0.672*** (0.142)	0.273*** (0.076)	0.399*** (0.095)
Observations	19000	19000	19000	16737	16737	16737
Adj. R-squared	0.380	0.377	0.338	0.861	0.818	0.823
Mean dep. var.	21.49	19.04	8.78	2.06	1.23	0.83

Notes: This table reports the results from a match sub-sample. Sample is restricted to treated municipalities (i.e., municipalities that contain a refined oil or gas pipeline) and control municipalities that exactly match on state and quintiles of pre-treatment (2015-2018) homicide rates per 100,000 residents, number of active cartels, formal employment share of the population, income level, and population. Matching weights from the Coarsened Exact Matching (CEM) procedure are included in event study regressions. Columns (1-3) show results on the homicide rates per per 100,000 inhabitants. Potential cartel-related homicides are males aged 18-40, killed by a firearm. Columns (4-6) report the results on the total number of active cartels. Dependent variable measures in Columns (5) and (6) measure presence on fuel-specialized cartels and non-fuel theft specialized cartels, respectively, where fuel-theft specialized cartels are identified from [Etelekt Consultores \(2016\)](#), [Langner \(2017\)](#), [Castillo \(2021\)](#), and [González \(2020\)](#). Municipality and year-fixed effects are included in all specifications. Clustered standard errors at the municipality level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A10: Log-Transformed Estimates: Crackdown Effects on Homicides

	All (1)	Male (2)	Female (3)	Pemex (4)	Cartel (5)	Military (6)
Gas pipeline $\times$ Post 2019	0.099** (0.047)	0.079* (0.048)	0.060* (0.033)	-0.002 (0.004)	0.125*** (0.047)	-0.002 (0.008)
Refined oil pipeline $\times$ Post 2019	0.021 (0.057)	0.015 (0.058)	0.027 (0.043)	-0.002 (0.007)	-0.044 (0.063)	0.007 (0.012)
Observations	19648	19648	19648	19648	19648	19648
Adj. R-squared	0.507	0.506	0.285	0.252	0.478	0.054
Mean dep. var.	1.96	1.84	0.54	0.01	1.11	0.01

Note: This table reports coefficient estimates and standard errors from Equation 8. All outcomes are normalized to homicide rates per 100,000 residents and then log-transformed. Homicides of Pemex employees are estimated based on homicides of individuals of working age (18–60 years old) insured through Pemex. Potential cartel-related homicides are classified as males aged 18–40 killed by a firearm. Homicides of military personnel are estimated based on individuals insured through SEDENA, ages 18–60, and killed by a firearm. Municipality and year fixed effects are included in all specifications. Standard errors clustered at the municipality level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Log-Transformed Estimates: Crackdown Effects on Cartel Presence

	Log(Number of Cartels)		
	All (1)	Huachicol Specialized (2)	Non-Huachicol Specialized (3)
Gas pipeline $\times$ Post 2019	0.0876*** (0.020)	0.0802*** (0.017)	0.0462*** (0.017)
Refined oil pipeline $\times$ Post 2019	0.0978*** (0.027)	0.0506** (0.024)	0.128*** (0.025)
Observations	17311	17311	17311
Adj. R-squared	0.821	0.781	0.793
Mean dep. var.	0.613	0.486	0.312

Note: This table reports coefficient estimates and standard errors from Equation 8. All dependent variables are log-transformed. In Column (1) the dependent variable measures total active cartels; in column (2) the dependent variable measures fuel theft-specialized cartels, and in column (3) the dependent variable measures non-fuel theft specialized cartels, where fuel-theft specialized cartels are identified from [Etellect Consultores \(2016\)](#), [Langner \(2017\)](#), [Castillo \(2021\)](#), and [González \(2020\)](#). Columns (4)–(6) report results for analogous binary categories to assess the extensive margin of cartel presence. Municipality and year-fixed effects are included in all specifications. Standard errors clustered at the municipality-level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

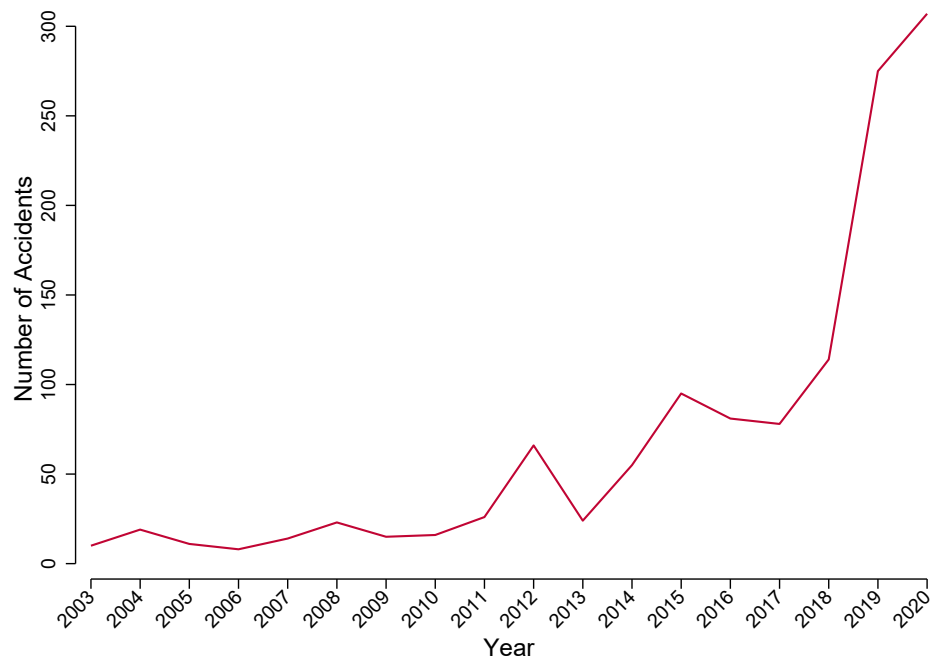
Table A12: Results Controlling for COVID

	Homicides			Number of Cartels		
	All	Male	Cartel	All	Huachicol Specialized	Non-Huachicol Specialized
	(1)	(2)	(3)	(4)	(5)	(6)
Gas pipeline $\times$ Post 2019	3.989*** (1.537)	3.441** (1.420)	2.082** (0.883)	0.363*** (0.090)	0.232*** (0.048)	0.131** (0.060)
Refined oil pipeline $\times$ Post 2019	3.829 (2.737)	3.496 (2.511)	1.150 (1.622)	0.958*** (0.155)	0.286*** (0.072)	0.672*** (0.118)
Observations	19648	19648	19648	17199	17199	17199
Adj. R-squared	0.356	0.355	0.325	0.884	0.819	0.867
Mean dep. var.	21.31	18.87	8.58	2.03	1.17	0.86

Notes: This table reports the results controlling for the reported number of covid cases per capita at the municipality-year level. Columns (1-3) show results on the homicide rates per per 100,000 inhabitants. Potential cartel-related homicides are males aged 18-40, killed by a firearm. Columns (4-6) report the results on the total number of active cartels. Dependent variable measures in Columns (5) and (6) measure presence on fuel-specialized cartels and non-fuel theft specialized cartels, respectively, where fuel-theft specialized cartels are identified from [Etellekt Consultores \(2016\)](#), [Langner \(2017\)](#), [Castillo \(2021\)](#), and [González \(2020\)](#). Municipality and year-fixed effects are included in all specifications. Clustered standard errors at the municipality level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

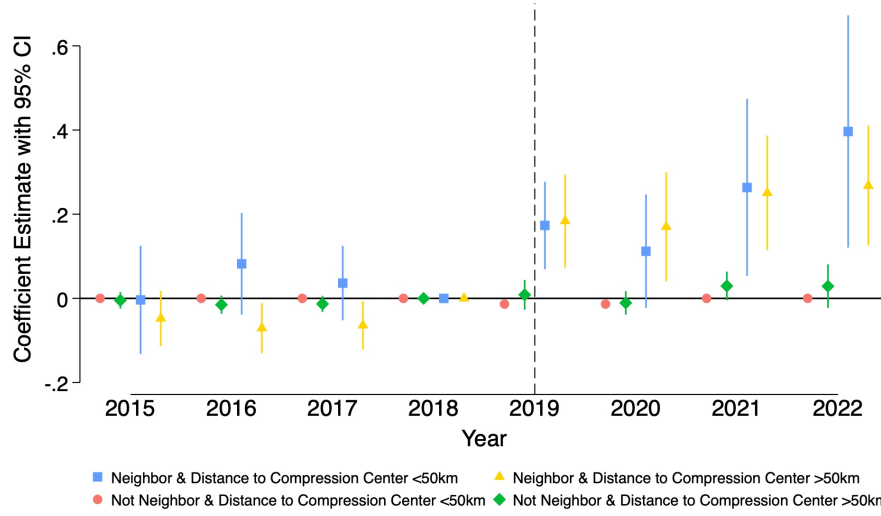
3 Supplementary Figures

Figure A1: Gas-Related Accidents



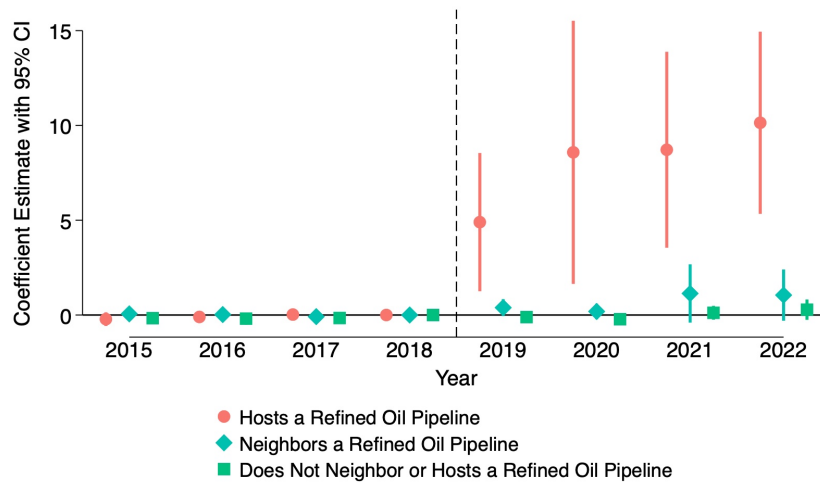
Note: Data are drawn from Mexico's National Center for Disaster Prevention (CE-NAPRED, 2021). Gas related accidents include fires, explosions, and other accidents.

Figure A2: Heterogeneity in Gas Pipeline Thefts, by Neighboring/Not-Neighboring an Oil Pipeline and Near/Far from a Gas Compression Center



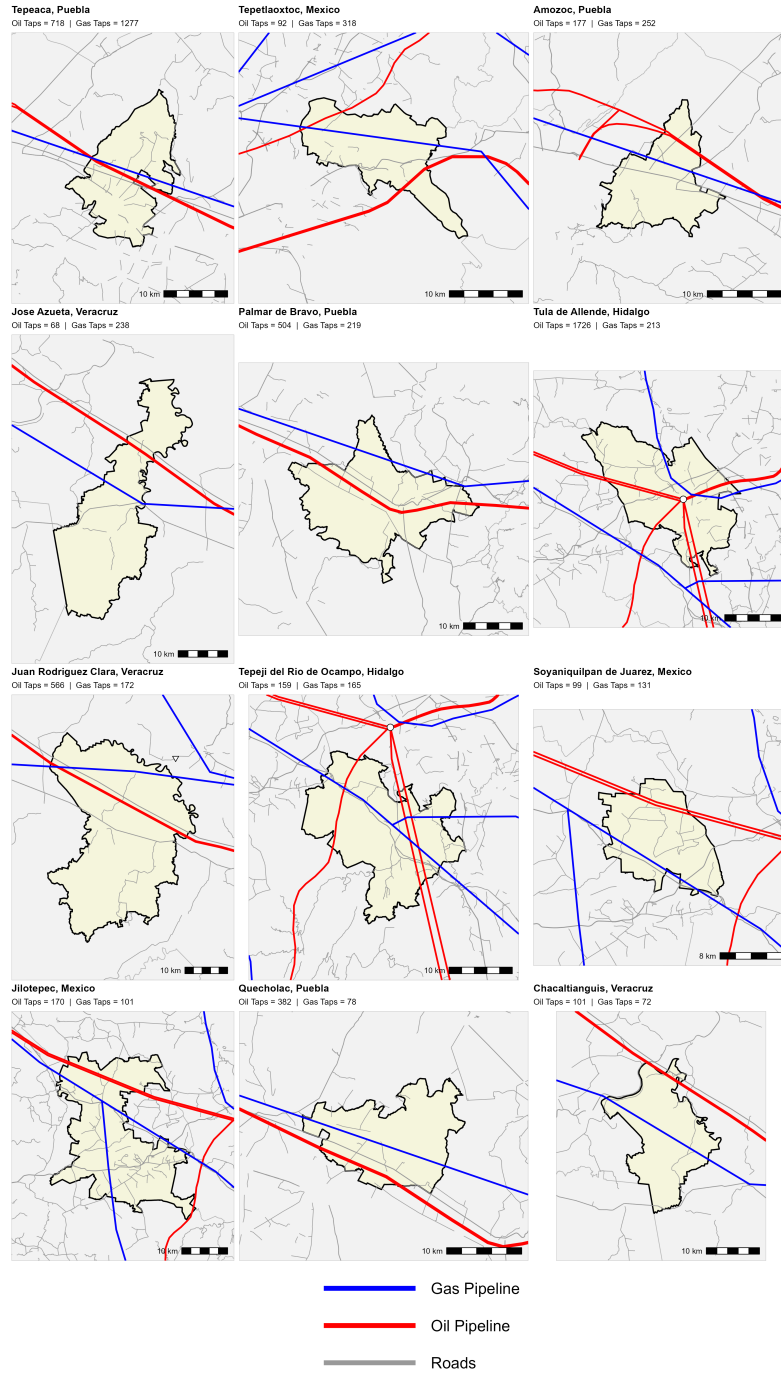
Note: Figure reports coefficient estimates and 95% confidence intervals for an event study specification with municipality and year fixed effects and four treatment interactions: neighbor/non-neighbor $\times$ near/far from a gas compression center. Near/far is defined using a 50km radius cutoff. Neighbors are defined as municipalities that are adjacent to or that themselves contain an oil pipeline. Standard errors are clustered at the municipality-level.

Figure A3: Dynamic Crackdown Effects on Gas Pipeline Thefts, by Oil Pipeline Proximity



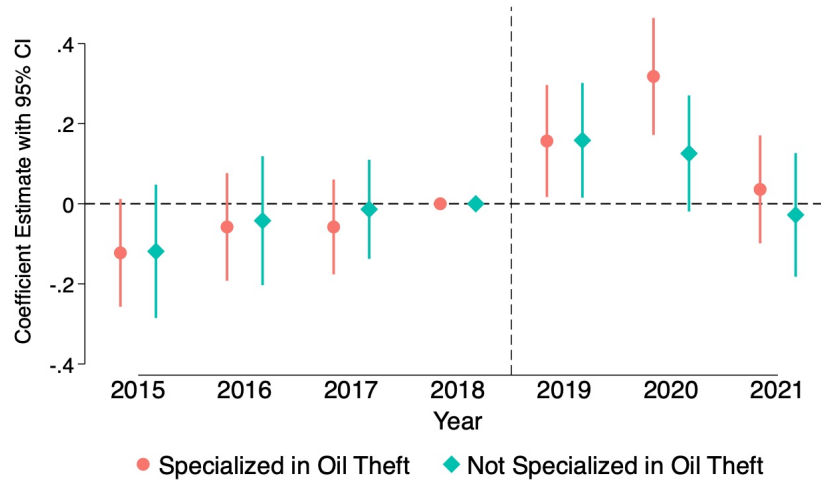
Note: Figure reports coefficient estimates and 95% confidence intervals for the number of illegal taps on gas pipelines, estimated separately for three sub-groups of gas pipeline municipalities: (i) those that also host a refined oil pipeline, (ii) those neighboring a municipality that hosts a refined oil pipeline, and (iii) those neither hosting nor neighboring a refined oil pipeline. Specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

Figure A4: Selected Municipalities with High Levels of Gas Pipeline Theft



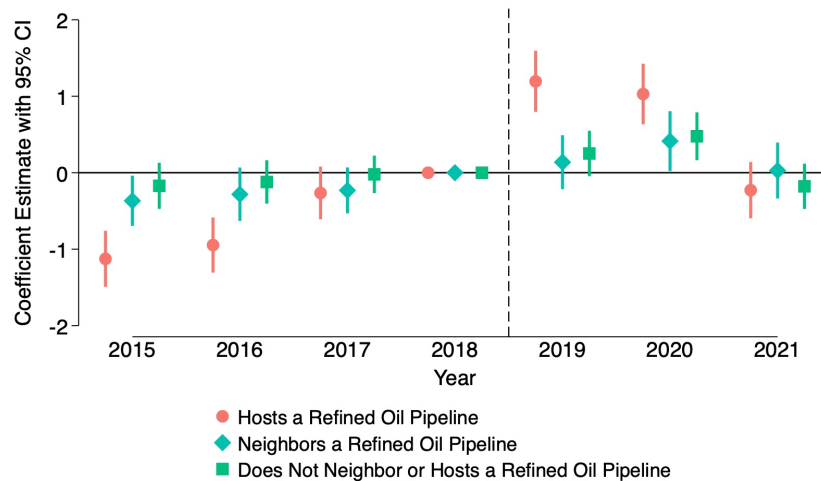
Note: Maps show selected municipalities with high levels of illegal gas pipeline taps that host both gas and oil pipeline infrastructure. Each panel highlights the local oil pipeline network, gas pipeline network, and road infrastructure within and surrounding the municipality.

Figure A5: Crackdown Effects on Cartel Presence in Gas Pipeline Municipalities, by Cartel Type



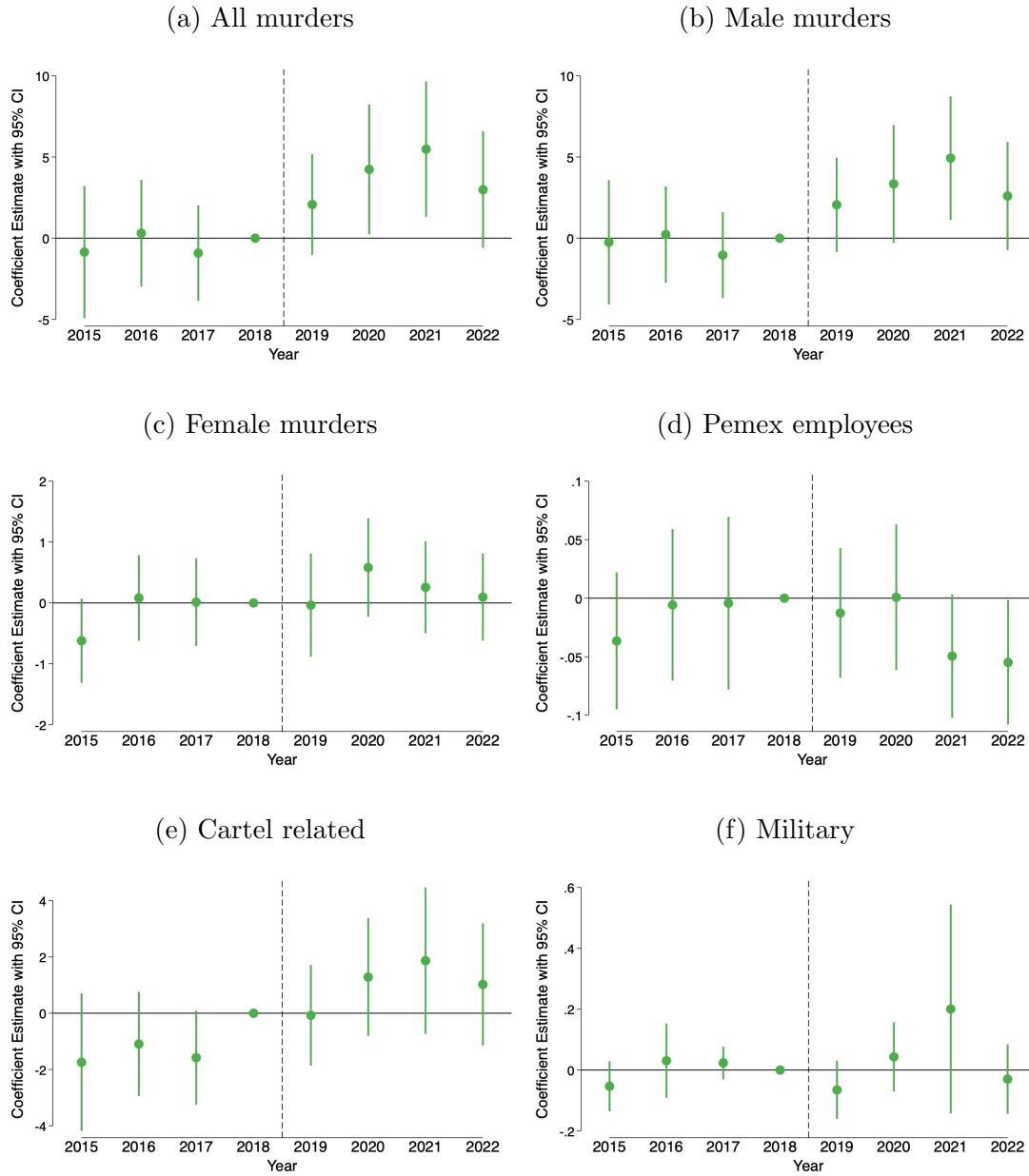
Note: Figure presents coefficient estimates and 95% confidence intervals from estimation of Equation 9, with outcomes disaggregated into the number of active cartels specialized in oil theft and other cartels. Treatment is defined as being a municipality with a GAS pipeline. Standard errors are clustered at the municipality-level. Fuel-theft-specialized cartels are identified from media and reports (Etellect Consultores, 2016; Langner, 2017; Castillo, 2021; González, 2020).

Figure A6: Dynamic Crackdown Effects on Cartel Presence in Gas Pipeline Municipalities, by Oil Pipeline Proximity



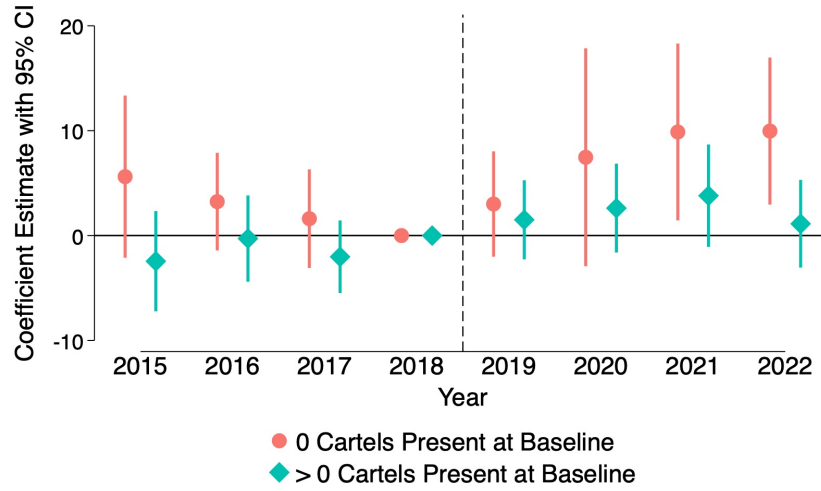
Note: Figure reports coefficient estimates and 95% confidence intervals for the number of active cartels in gas pipeline municipalities, estimated separately for three sub-groups: municipalities that also host a refined oil pipeline, municipalities neighboring a refined oil pipeline municipality, and municipalities neither hosting nor neighboring a refined oil pipeline. Specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

Figure A7: Crackdown Effects on Homicides in Municipalities with Gas Pipelines (Disaggregated)



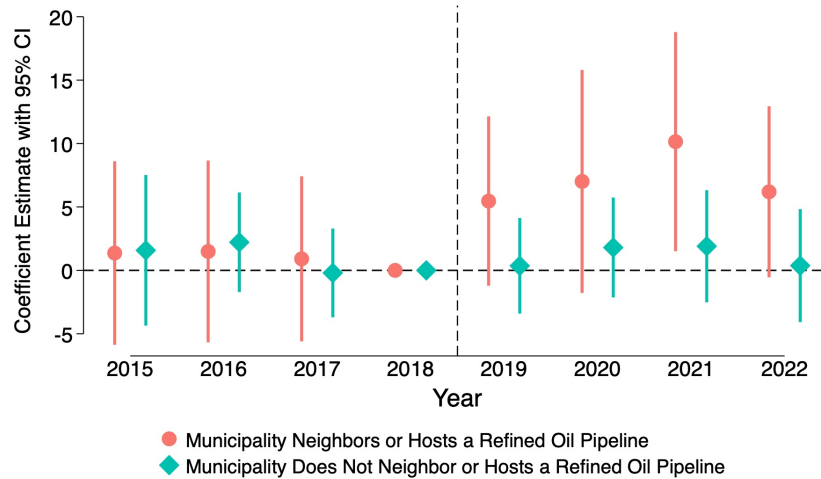
Note: Figure reports coefficient estimates and 95% confidence intervals for homicide rates per 100,000 residents among (a) all residents, (b) males, (c) females, (d) Pemex employees, (e) likely cartel-related murders, and (f) military personnel, estimated using Equation 9. Specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

Figure A8: Dynamic Crackdown Effects on Homicides in Gas Pipeline Municipalities, by Baseline Cartel Presence



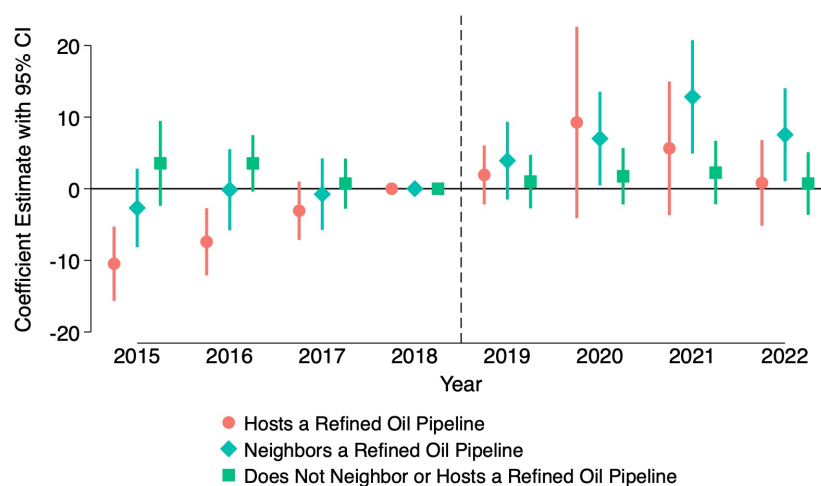
Note: Figure reports coefficient estimates and 95% confidence intervals for homicide rates in gas pipeline municipalities relative to non-pipeline municipalities, estimated separately for municipalities with and without any recorded cartel presence during the 2015–2018 baseline period. Specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

Figure A9: Dynamic Crackdown Effects on Homicides in Gas Pipeline Municipalities, Near/Far from Oil Pipelines



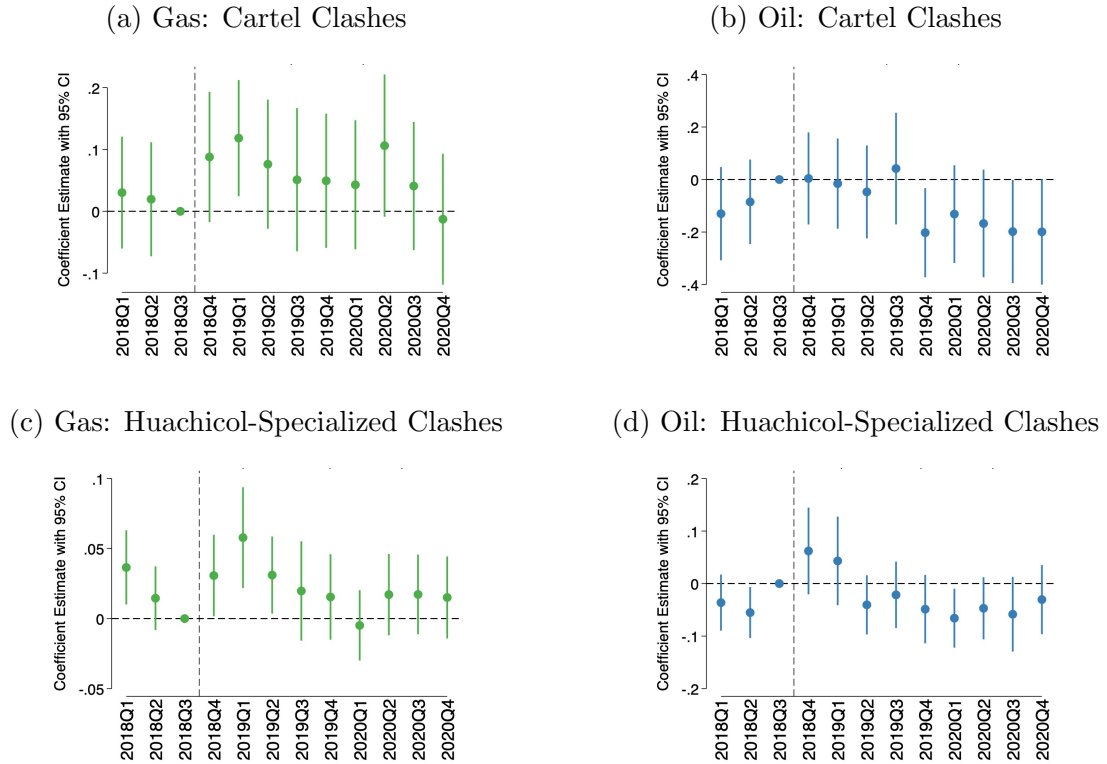
Note: Figure reports coefficient estimates and 95% confidence intervals for homicide rates reported in municipalities hosting gas pipelines, decomposed into two groups: (i) municipalities that neighbor or host a refined oil pipeline as well as a gas pipeline, and (ii) municipalities that do not neighbor or host a refined oil pipeline. Specification includes municipality and year fixed effects, and standard errors are clustered at the municipality level.

Figure A10: Dynamic Crackdown Effects on Homicides in Gas Pipeline Municipalities, by Oil Pipeline Proximity



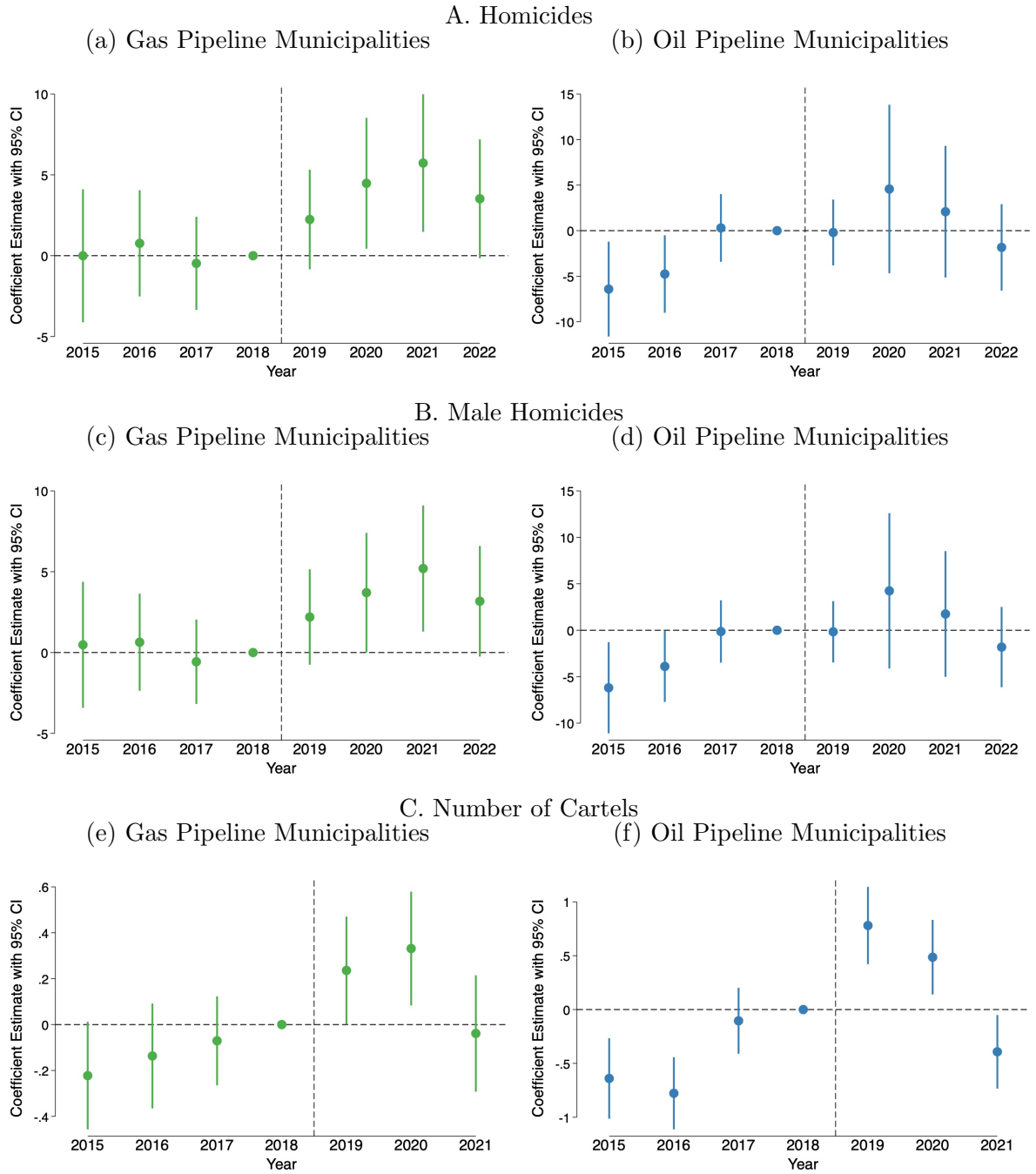
Note: Figure reports coefficient estimates and 95% confidence intervals for homicide rates per 100,000 inhabitants in gas pipeline municipalities, estimated separately for three sub-groups: municipalities that also host a refined oil pipeline, municipalities neighboring a refined oil pipeline municipality, and municipalities neither hosting nor neighboring a refined oil pipeline. Specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

Figure A11: Dynamic Crackdown Effects on Violent Conflict (ACLED)



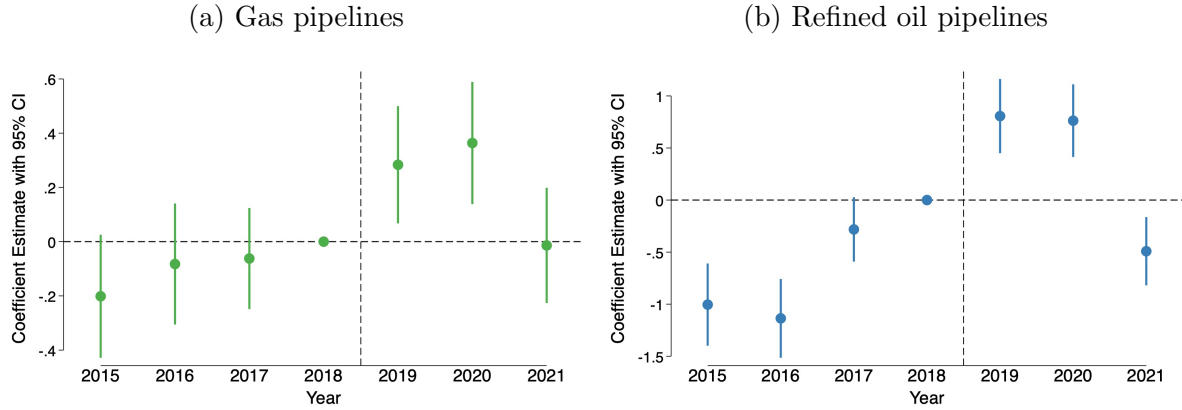
Note: Figure reports coefficient estimates and 95% confidence intervals for cartel armed clashes (panels a–b) and huachicol-specialized clashes (panels c–d) in gas pipeline municipalities (panels a, c) and refined oil pipeline municipalities (panels b, d), relative to non-pipeline municipalities. Events are recorded by the Armed Conflict Location and Event Data Project (ACLED). Specification includes municipality and quarter fixed effects. Standard errors are clustered at the municipality level. Huachicol-specialized clashes are those involving huachicol-specialized cartels (Cartel Jalisco Nueva Generación, Los Zetas, the Sinaloa Cartel, the Beltrán Leyva Organization, La Familia Michoacana, Los Caballeros Templarios, and the Cartel de Santa Rosa de Lima) as well as clashes specifically classified as huachicol-related by ACLED.

Figure A12: Event Studies with Matched Sub-sample



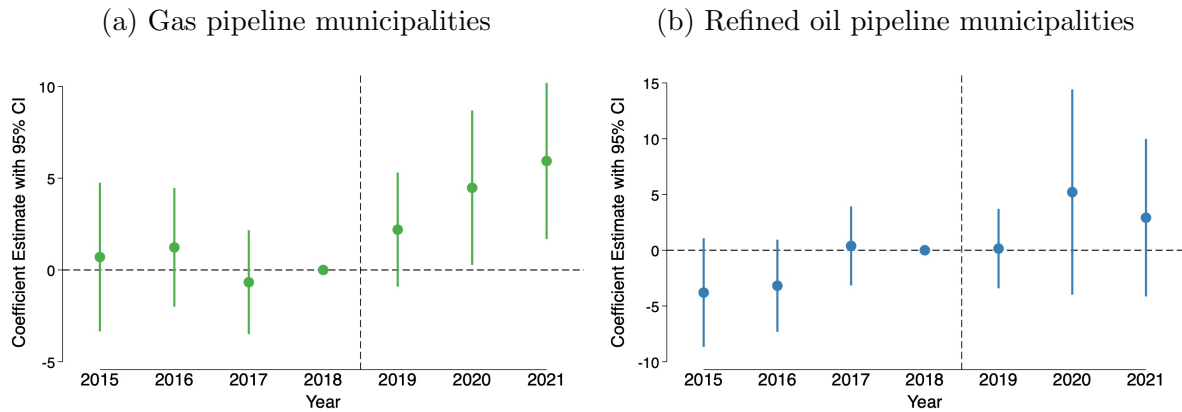
Note: Event studies are estimated analogously to Equation 9, including year and municipality fixed effects and clustering standard errors at the municipality level. Panel A show results on all homicides, Panel B presents results for homicides of males, and Panel C show results on the number of cartels. Outcomes in Panels A and B are defined as total homicides per 100,000 residents. Outcomes in Panel C are defined as the number of active cartels in a municipality. Sample is restricted to treated municipalities (i.e., municipalities that contain a refined oil or gas pipeline) and control municipalities that exactly match on state and quintiles of pre-treatment (2015-2018) homicide rates per 100,000 residents, number of active cartels, formal employment share of the population, income level, and population. Matching weights from the Coarsened Exact Matching (CEM) procedure are included in event study regressions.

Figure A13: Dynamic Crackdown Effects on Cartel Presence, Excluding CJNG



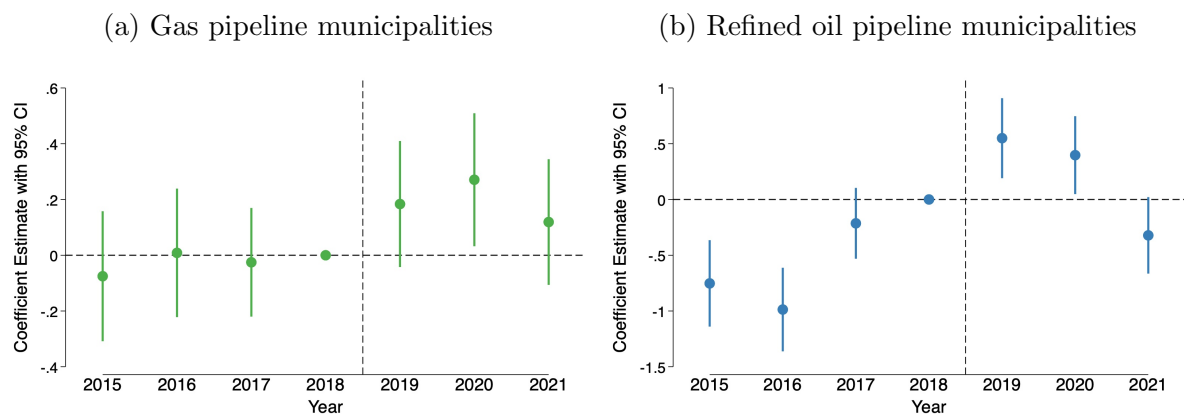
Note: Figure reports coefficient estimates and 95% confidence intervals for the number of active cartels, excluding CJNG, in (a) municipalities hosting gas pipelines and (b) municipalities hosting refined oil pipelines, relative to non-pipeline municipalities. Specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

Figure A14: Dynamic Crackdown Effects on Homicides, Controlling for Baseline CJNG Presence



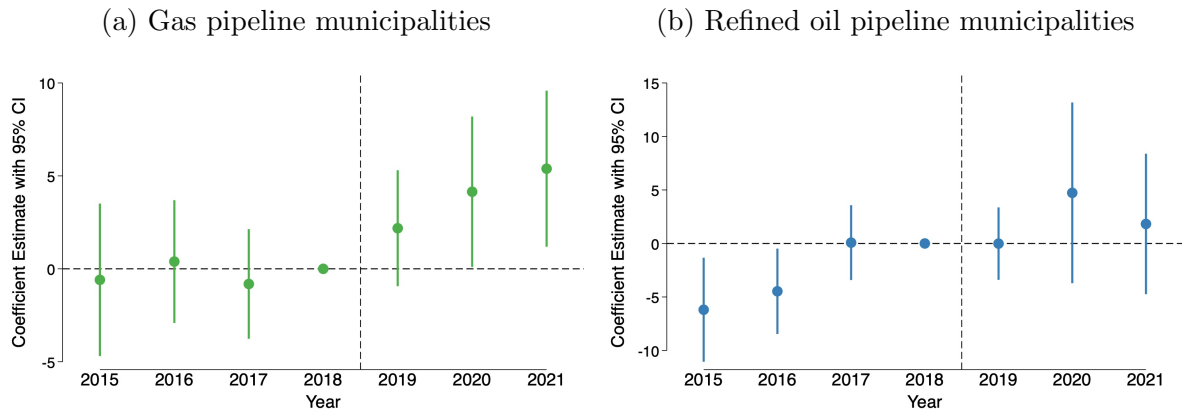
Note: Figure reports coefficient estimates and 95% confidence intervals for homicide rates per 100,000 inhabitants in (a) gas pipeline municipalities and (b) refined oil pipeline municipalities, relative to non-pipeline municipalities. This specification controls for baseline CJNG presence, defined as any recorded CJNG activity during 2015-2018, interacted with year indicators. Specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

Figure A15: Dynamic Crackdown Effects on Cartel Presence, Controlling for Baseline CJNG Presence



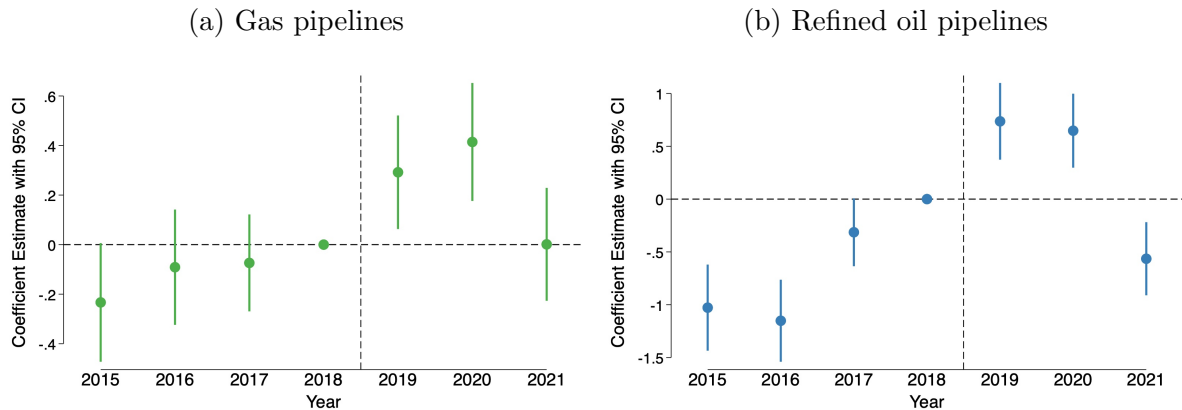
Note: Figure reports coefficient estimates and 95% confidence intervals for the number of active cartels in (a) gas pipeline municipalities and (b) refined oil pipeline municipalities, relative to non-pipeline municipalities. This specification controls for baseline CJNG presence, defined as any recorded CJNG activity during 2015-2018, interacted with year indicators. Specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

Figure A16: Dynamic Crackdown Effects on Homicides, Excluding Municipalities with Fentanyl Seizures (2015-2018)



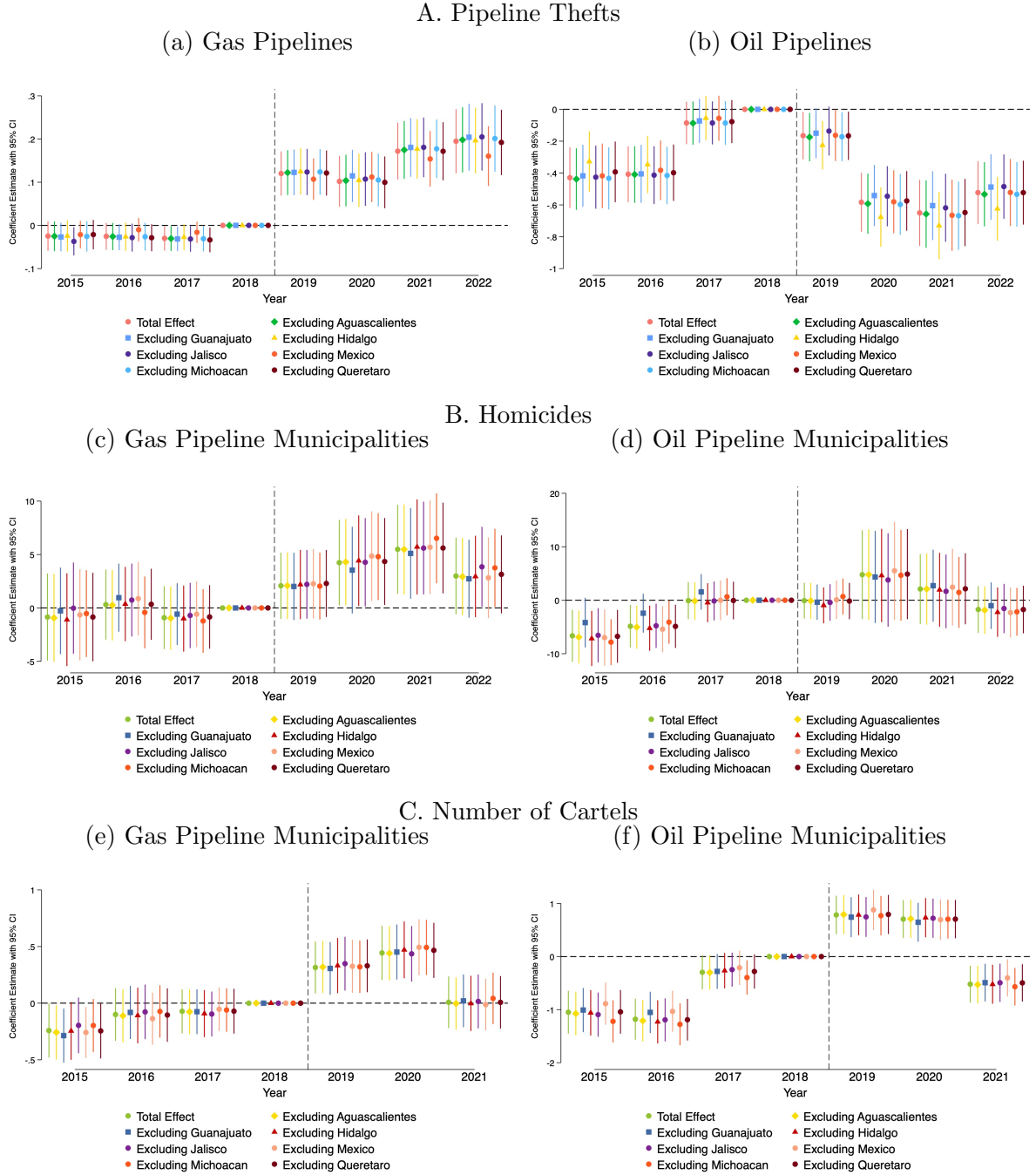
Note: Figure reports coefficient estimates and 95% confidence intervals for homicide rates per 100,000 inhabitants in (a) gas pipeline municipalities and (b) refined oil pipeline municipalities, relative to non-pipeline municipalities. This specification controls for baseline CJNG presence, defined as any recorded CJNG activity during 2015-2018, interacted with year indicators. Specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

Figure A17: Dynamic Crackdown Effects on Cartel Presence, Excluding Municipalities with Fentanyl Seizures (2015-2018)



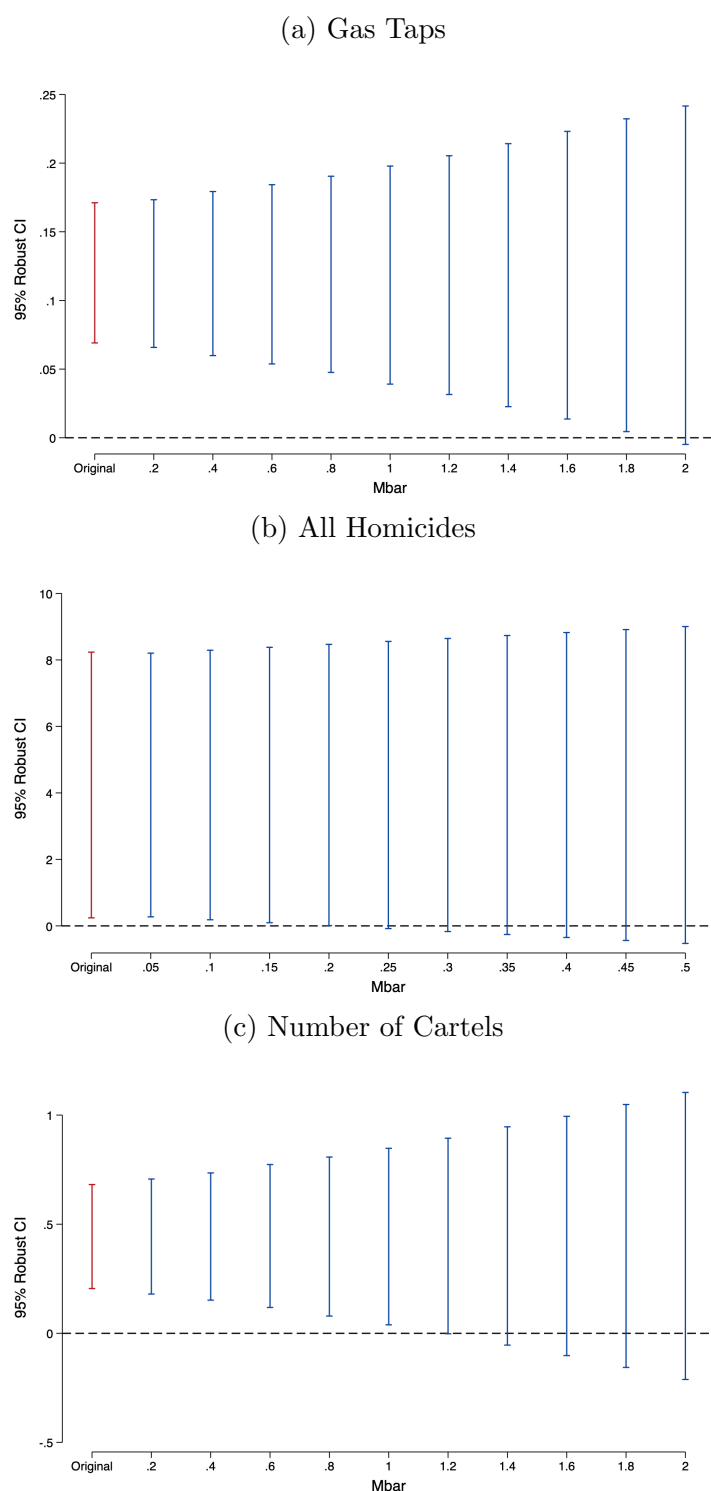
Note: Figure reports coefficient estimates and 95% confidence intervals for the number of active cartels, excluding municipalities in which Fentanyl was seized anytime between 2015-2018, in (a) municipalities hosting gas pipelines and (b) municipalities hosting refined oil pipelines, relative to non-pipeline municipalities. Specification includes municipality and year fixed effects. Standard errors are clustered at the municipality level.

Figure A18: Leave-One-Out Analysis: Event Studies Excluding Targeted States



Note: Event studies are estimated analogously to Equation 9, including year and municipality fixed effects and clustering standard errors at the municipality level. However, these specifications are estimated repeatedly, leaving out one crackdown-affected state each time to assess stability of results to potential outlier states.

Figure A19: [Rambachan and Roth \(2023\)](#) Breakeven Analysis for Key Outcomes in Gas Pipeline Municipalities



Note: Panels report robust confidence sets for different values of $\bar{M}$, based on the *honestdid* package from [Rambachan and Roth \(2023\)](#), for main results (a) gas taps, (b) homicide rates, and (c) cartel presence. “Breakeven” $\bar{M}$ values indicate the magnitude of the deviation of a linear extrapolation of post-trends from pre-trend violations under which the null hypothesis can still be rejected.